

A Novel Finite-Sum Inequality-Based Method for Robust H_∞ Control of Uncertain Discrete-Time Takagi–Sugeno Fuzzy Systems With Interval-Like Time-Varying Delays

Xian-Ming Zhang, *Member, IEEE*, Qing-Long Han, *Senior Member, IEEE*, and Xiaohua Ge

Abstract—This paper is concerned with the problem of robust H_∞ control of an uncertain discrete-time Takagi–Sugeno fuzzy system with an interval-like time-varying delay. A novel finite-sum inequality-based method is proposed to provide a tighter estimation on the forward difference of certain Lyapunov functional, leading to a less conservative result. First, an auxiliary vector function is used to establish two finite-sum inequalities, which can produce tighter bounds for the finite-sum terms appearing in the forward difference of the Lyapunov functional. Second, a matrix-based quadratic convex approach is employed to equivalently convert the original matrix inequality including a quadratic polynomial on the time-varying delay into two boundary matrix inequalities, which delivers a less conservative bounded real lemma (BRL) for the resultant closed-loop system. Third, based on the BRL, a novel sufficient condition on the existence of suitable robust H_∞ fuzzy controllers is derived. Finally, two numerical examples and a computer-simulated truck-trailer system are provided to show the effectiveness of the obtained results.

Index Terms—Finite-sum inequalities, robust H_∞ fuzzy control, Takagi–Sugeno (T–S) fuzzy systems, time-varying delays.

I. INTRODUCTION

TAKAGI–SUGENO (T–S) fuzzy models provide an effective method to approximate a class of complex nonlinear systems [1]. For example, truck trailer systems, stirred tank reactor systems, and chaotic Lorenz systems can be represented as certain T–S fuzzy model [2]. A conspicuous feature of a T–S fuzzy model is that a family of IF–THEN rules are defined to describe local linear input–output relations of a nonlinear system, and the overall system can be obtained by smoothly mixing those local linear models through membership functions. Consequently, fruitful linear system theory can be applied to the theoretical analysis and design of T–S

fuzzy systems [3]–[8]. On the other hand, time-delays are frequently encountered in practical control systems, and their existence is commonly regarded as a source of system performance degradation [9]–[12]. During the past two decades, a large number of results on T–S fuzzy systems with time-delays have been reported in the literature, one can refer to [13]–[18] and references therein.

H_∞ control is a fundamental issue in the field of control [19], aiming to design controllers achieving stabilization with guaranteed performance [20], [21]. The main feature of using the H_∞ method is that the control problem is expressed as a mathematical optimization problem that is solved by seeking a suitable controller. Compared with some classical control techniques, H_∞ method is readily applicable to problems involving multivariate systems with cross-coupling between channels [22]. For T–S fuzzy systems with time-varying delays, the objective of H_∞ control is to design an optimal fuzzy controller such that in the worst case of external noises the resultant closed-loop system is asymptotically stable with a prescribed H_∞ disturbance attenuation level $\gamma > 0$. For a given upper bound of a time-varying delay, the smaller the γ is obtained, the better the performance of the corresponding closed-loop system is achieved. Thus, the obtained minimum γ is regarded as a key index to measure the conservatism of a bounded real lemma (BRL) [23]–[25].

For discrete-time T–S fuzzy systems with time-varying delays, the Lyapunov functional method is a powerful tool for studying the H_∞ control problem [26]. In order to derive some less conservative results, two important issues should be taken into account. One issue is to choose a *proper* Lyapunov functional. It has been shown that a fuzzy-basis-dependent (FBD) Lyapunov functional can produce less conservative results [27], [28]. Thus, a number of results on H_∞ control have been derived based on FBD Lyapunov functionals (see [25], [26], [29], [30]). The other issue is to provide a tighter estimation for the forward difference of the chosen FBD Lyapunov functional. There usually exists a finite-sum term as

$$\mathcal{F}(\tau(k)) := - \sum_{j=k-d_2}^{k-\tau(k)-1} \eta^T(j) R \eta(j) \quad (1)$$

in the forward difference of a certain FBD Lyapunov functional, where the time-varying delay $\tau(k)$ satisfies $\tau(k) \leq d_2$

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The authors are with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC 3122, Australia (e-mail: xianmingzhang@swin.edu.au; qhan@swin.edu.au; xge@swin.edu.au).

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with d_2 being an integer; $R = R^T > 0$ and $\eta(k) = x(k+1) - x(k)$ with $x(k)$ being the system state. It is clear that a tighter upper bound of the term $\mathcal{F}(\tau(k))$ can provide a tighter estimation on the forward difference of the FBD Lyapunov functional, leading to a less conservative BRL. Recalling some existing approaches, the term $\mathcal{F}(\tau(k))$ is usually bounded using the free-weighting matrix technique [25], [30] or the Jensen-type inequality. As shown in [31], applying the free-weighting matrix technique and the Jensen-type inequality to the term $\mathcal{F}(\tau(k))$ yields the same minimum upper bound, which gives

$$\mathcal{F}(\tau(k)) \leq -\frac{1}{d_2-\tau(k)} \mathfrak{S}_1^T R \mathfrak{S}_1 \quad (2)$$

where $\mathfrak{S}_1 := x(k - \tau(k)) - x(k - d_2)$. Fortunately, the above inequality (2) is improved in [31] and [32], which reads as

$$\mathcal{F}(\tau(k)) \leq -\frac{1}{d_2-\tau(k)} \mathfrak{S}_1^T R \mathfrak{S}_1 - \frac{3}{d_2-\tau(k)} \mathfrak{S}_2^T R \mathfrak{S}_2 + \mathcal{B}_1 \quad (3)$$

where $\mathfrak{S}_2 := x(k - \tau(k)) + x(k - d_2) - [2/(d_2 - \tau(k) + 1)] \times \sum_{j=k-d_2}^{k-\tau(k)} x(j)$, and

$$\mathcal{B}_1 := -\frac{6}{(d_2-\tau(k))(d_2-\tau(k)-1)} \mathfrak{S}_2^T R \mathfrak{S}_2. \quad (4)$$

Note that $\mathfrak{S}_2 = 0$ for both $\tau(k) = d_2$ and $\tau(k) = d_2 - 1$. Thus, one can see clearly that the inequality (3) provides a smaller upper bound than (2) for the term $\mathcal{F}(\tau(k))$ due to the fact that $-[3/(d_2 - \tau(k))] \mathfrak{S}_2^T R \mathfrak{S}_2 + \mathcal{B}_1 \leq 0$ for $\tau(k) \leq d_2$. In [32], the inequality (3) is employed to investigate the stability of linear discrete-time systems with time-varying delay. However, the term $\mathcal{B}_1$ in (4) is *overly* bounded by zero. If not doing so, it is difficult to cope with the term $\mathcal{B}_1$, since a quadratic polynomial on the time-varying delay $\tau(k)$ appears in the form of its inverse. Therefore, two natural questions arise.

- 1) Is there any tighter upper bound than (3) for $\mathcal{F}(\tau(k))$?
- 2) How should one deal with the term $\mathcal{B}_1$ so that less conservative results can be obtained?

Answering these questions motivates this paper.

This paper is concerned with the problem of robust H_∞ control for a class of discrete-time T-S fuzzy systems with interval-like time-varying delay. The main contributions of this paper lie in two aspects.

- 1) A tighter upper bound than (3) is derived for $\mathcal{F}(\tau(k))$, which is given by

$$\mathcal{F}(\tau(k)) \leq -\frac{1}{d_2-\tau(k)} \mathfrak{S}_1^T R \mathfrak{S}_1 - \frac{3}{d_2-\tau(k)} \mathfrak{S}_2^T R \mathfrak{S}_2 - \frac{5}{d_2-\tau(k)} \mathfrak{S}_3^T R \mathfrak{S}_3 + \mathcal{B}_1 + \mathcal{B}_2 \quad (5)$$

where $\mathfrak{S}_3 := \mathfrak{S}_1 + \sum_{j=k-d_2}^{k-\tau(k)} [(6(2k-d_2-\tau(k)-2j))/((d_2-\tau(k)+1)(d_2-\tau(k)+2))]x(j)$ and

$$\mathcal{B}_2 := -\frac{10}{(d_2-\tau(k))(d_2-\tau(k)-1)} \mathfrak{S}_3^T R \mathfrak{S}_3. \quad (6)$$

Since $\mathfrak{S}_3 = 0$ if $\tau(k) \in \{d_2, d_2 - 1\}$, the upper bound of $\mathcal{F}(\tau(k))$ in (5) is tighter than that in (3) due to the fact that $-[5/(d_2 - \tau(k))] \mathfrak{S}_3^T R \mathfrak{S}_3 + \mathcal{B}_2 \leq 0$ for $\tau(k) \leq d_2$. Moreover, in a similar way to the proof of (5), a novel inequality for double-sum terms is also established.

- 2) A novel method is presented to deal with the terms $\mathcal{B}_1$ and $\mathcal{B}_2$, which are not necessary to be overly bounded by zero. First, using Lemma 2 yields an estimation on $\mathcal{B}_1$ (or $\mathcal{B}_2$), where the quadratic polynomial

$(d_2 - \tau(k))(d_2 - \tau(k) - 1)$ on $\tau(k)$ appears no longer in the form of its inverse. Second, the idea of a matrix-based quadratic convex approach [33] is then introduced to produce a less conservative BRL for the resultant closed-loop T-S fuzzy systems, which is demonstrated through a computer-simulated truck-trailer system.

Since the proposed method is closely related to two novel finite-sum inequalities, we refer to it as a novel finite-sum inequality-based method.

This paper is organized as follows. The problem formulation is stated in Section II. Two novel finite-sum inequalities are established in Section III. Robust H_∞ performance analysis is made in Section IV. The design of robust H_∞ controllers is presented in Section V. Two examples are given in Section VI to demonstrate the effectiveness of the proposed method. The conclusion is made in Section VII.

Notations: The notations in this paper are the same as [34]. The space of square-summable vector sequences is denoted by l_2 , and for $w(k) \in l_2$ its norm is given by $\|w\|_2 = [\sum_{k=0}^{\infty} w^T(k)w(k)]^{1/2}$. The expression $\text{He}\{A\}$ means $A + A^T$. The symbol “ $\star$ ” represents the symmetric term in a symmetric matrix.

II. PROBLEM DESCRIPTION

Consider the uncertain discrete-time T-S fuzzy model with an interval-like time-varying delay, in which the plant rules are described as follows.

Plant Rule i: IF $\theta_1(k)$ is μ_{i1} , $\theta_2(k)$ is μ_{i2} , ..., and $\theta_s(k)$ is μ_{is} , THEN

$$\begin{cases} x(k+1) = \tilde{A}_i x(k) + \tilde{A}_{di} x(k - \tau(k)) + B_{1i} u(k) + B_{2i} w(k) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2 + 1, \dots, 0\} \\ z(k) = \tilde{L}_i x(k) + \tilde{L}_{di} x(k - \tau(k)) + B_{3i} w(k) \end{cases} \quad (7)$$

where $i \in \mathcal{S} := \{1, 2, \dots, r\}$ with r being the number of IF-THEN rules; μ_{ij} is the fuzzy set; $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^m$, and $z(k) \in \mathbb{R}^q$ are the system state, control input, and controlled output, respectively; $w(k) \in \mathbb{R}^q$ is the disturbance input belonging to l_2 ; $\theta(k) := \text{col}\{\theta_1(k), \theta_2(k), \dots, \theta_s(k)\}$ are the known premise variables that may be functions of state variables, external disturbance and/or time [35]; and $\phi(k)$ is the initial condition. The coefficient matrices B_{1i} , B_{2i} , and B_{3i} are constant real matrices and $\tilde{A}_i$, $\tilde{A}_{di}$, and $\tilde{L}_i$ and $\tilde{L}_{di}$ are given by

$$\begin{aligned} \tilde{A}_i &:= A_i + D_{1i} F_i(k) E_{1i}, & \tilde{A}_{di} &:= A_{di} + D_{1i} F_i(k) E_{2i} \\ \tilde{L}_i &:= L_i + D_{2i} F_i(k) E_{1i}, & \tilde{L}_{di} &:= L_{di} + D_{2i} F_i(k) E_{2i} \end{aligned}$$

where A_i , A_{di} , L_i , L_{di} , D_{1i} , D_{2i} , E_{1i} , and E_{2i} are constant real matrices and $F_i(k)$ is an unknown time-varying matrix satisfying

$$F_i^T(k) F_i(k) \leq I, \quad \forall k. \quad (8)$$

The delay $\tau(k)$ is an interval-like time-varying delay satisfying

$$d_1 \leq \tau(k) \leq d_2 \quad (9)$$

where d_1 and d_2 are two positive integers.

In order to avoid a complicated defuzzification process of fuzzy controllers, the premise variables are assumed to be

independent of the control input $u(k)$ and the disturbance $w(k)$. Thus, for a given pair of $(x(k), u(k))$, the output of the fuzzy system (7) reads as

$$\begin{aligned} x(k+1) &= \sum_{i=1}^r h_i(\theta(k)) \left\{ \tilde{A}_i x(k) + \tilde{A}_{di} x(k-\tau(k)) \right. \\ &\quad \left. + B_{1i} u(k) + B_{2i} w(k) \right\} \\ z(k) &= \sum_{i=1}^r h_i(\theta(k)) \left\{ \tilde{L}_i x(k) + \tilde{L}_{di} x(k-\tau(k)) + B_{3i} w(k) \right\} \end{aligned}$$

where $h_i(\theta(k))$ ($i \in \mathcal{S}$) are the fuzzy basis functions given by

$$h_i(\theta(k)) = \frac{\mu_i(\theta(k))}{\sum_{j=1}^r \mu_j(\theta(k))}, \quad \mu_i(\theta(k)) := \prod_{j=1}^s \mu_{ij}(\theta_j(k))$$

with $\mu_{ij}(\theta_j(k))$ being the grade of the membership of $\theta_j(k)$ in μ_{ij} . Suppose that $\mu_j(\theta(k)) \geq 0$ ($i \in \mathcal{S}$) and $\sum_{i=1}^r \mu_i(\theta(k)) > 0$ for $k \geq 0$. Hence, it is clear that $h_i(\theta(k)) \geq 0$ and $\sum_{i=1}^r h_i(\theta(k)) = 1$.

In this paper, we are interested in designing a state feedback fuzzy controller, whose control rule is described as follows.

Control Rule i : IF $\theta_1(k)$ is μ_{i1} , $\theta_2(k)$ is μ_{i2} , ..., and $\theta_s(k)$ is μ_{is} , THEN

$$u(k) = K_i x(k) \quad (10)$$

where K_i ($i \in \mathcal{S}$) are the gain matrices to be determined. With the fuzzy controller, the closed-loop system is given by

$$\begin{cases} x(k+1) = \mathcal{A}(k)x(k) + \mathcal{A}_d(k)x(k-\tau(k)) + \mathcal{B}_1(k)w(k) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2+1, \dots, 0\} \\ z(k) = \mathcal{L}(k)x(k) + \mathcal{L}_d(k)x(k-\tau(k)) + \mathcal{B}_2(k)w(k) \end{cases} \quad (11)$$

where

$$\begin{aligned} \mathcal{A}(k) &:= \sum_{i=1}^r \sum_{j=1}^r h_i(\theta(k)) h_j(\theta(k)) [A_i + B_{1i} K_j + D_{1i} F_i(k) E_{1i}] \\ \mathcal{A}_d(k) &:= \sum_{i=1}^r h_i(\theta(k)) [A_{di} + D_{1i} F_i(k) E_{2i}] \\ \mathcal{L}(k) &:= \sum_{i=1}^r h_i(\theta(k)) [L_i + D_{2i} F_i(k) E_{1i}] \\ \mathcal{L}_d(k) &:= \sum_{i=1}^r h_i(\theta(k)) [L_{di} + D_{2i} F_i(k) E_{2i}] \\ \mathcal{B}_1(k) &:= \sum_{i=1}^r h_i(\theta(k)) B_{2i}, \quad \mathcal{B}_2(k) := \sum_{i=1}^r h_i(\theta(k)) B_{3i}. \end{aligned}$$

The problem to be addressed in this paper is stated as follows.

Robust H_∞ Fuzzy Control: For given integers d_1 and d_2 satisfying $0 < d_1 < d_2$ and a scalar $\gamma > 0$, design controller gains K_i ($i \in \mathcal{S}$) such that:

1) the closed-loop system (11) with $w(k) \equiv 0$, that is

$$\begin{cases} x(k+1) = \mathcal{A}(k)x(k) + \mathcal{A}_d(k)x(k-\tau(k)) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2+1, \dots, 0\} \end{cases} \quad (12)$$

is robustly asymptotically stable;

2) the closed-loop system (11) achieves a prescribed H_∞ level $\gamma > 0$. That is, for nonzero $w(k) \in l_2$, the inequality $\|z\|_2 < \gamma \|w\|_2$ is satisfied under zero initial conditions.

In this paper, we propose a novel finite-sum inequality-based method to solve the above problem of robust H_∞ fuzzy control. As part of the finite-sum inequality-based method, we apply the idea of a matrix-based quadratic convex approach to formulate a less conservative BRL for the resultant closed-loop T-S fuzzy systems. For this purpose, two useful lemmas are first introduced as follows.

Lemma 1 [33]: Let $\mathcal{X}_0$, $\mathcal{X}_1$, and $\mathcal{X}_2$ be $m \times m$ real symmetric matrices and a continuous function τ satisfy $\tau_1 \leq \tau \leq \tau_2$, where τ_1 and τ_2 are constants satisfying $0 \leq \tau_1 \leq \tau_2$. If $\mathcal{X}_0 \geq 0$, then

$$\begin{aligned} \tau^2 \mathcal{X}_0 + \tau \mathcal{X}_1 + \mathcal{X}_2 &< 0 \quad (\leq 0), \quad \forall \tau \in [\tau_1, \tau_2] \\ \iff \tau_i^2 \mathcal{X}_0 + \tau_i \mathcal{X}_1 + \mathcal{X}_2 &< 0 \quad (\leq 0), \quad (i = 1, 2). \end{aligned}$$

Lemma 2 [36]: Let ξ_1 and ξ_2 be real column vectors with dimensions of n_1 and n_2 , respectively. For given real positive symmetric matrices $M_1 \in \mathbb{R}^{n_1 \times n_1}$ and $M_2 \in \mathbb{R}^{n_2 \times n_2}$, if $\begin{bmatrix} M_1 & S \\ S^T & M_2 \end{bmatrix} \geq 0$, then the following inequality holds for any scalar $\kappa > 0$ and matrix $S \in \mathbb{R}^{n_1 \times n_2}$:

$$-2\xi_1^T S \xi_2 \leq \kappa \xi_1^T M_1 \xi_1 + \kappa^{-1} \xi_2^T M_2 \xi_2.$$

III. TWO NOVEL FINITE-SUM INEQUALITIES

In this section, we first introduce two novel finite-sum inequalities, which are useful in the H_∞ performance analysis for the system (11).

Theorem 1: For a given positive definite matrix $R \in \mathbb{R}^{n \times n}$, integers r_1 and r_2 with $r_2 > r_1$, and an n -dimensional real vector sequence $\{x(r_2), \dots, x(r_1)\}$, the following two inequalities hold:

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \eta^T(j) R \eta(j) &\geq \frac{1}{\bar{h}} \varpi_0^T R \varpi_0 + \frac{3(\bar{h}+1)}{\bar{h}(\bar{h}-1)} \varpi_1^T R \varpi_1 \\ &\quad + \frac{5(\bar{h}+1)(\bar{h}+2)}{\bar{h}(\bar{h}-1)(\bar{h}-2)} \varpi_2^T R \varpi_2 \end{aligned} \quad (13)$$

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} (r_2-j) \eta^T(j) R \eta(j) &\geq \frac{\bar{h}+1}{2\bar{h}} (\varpi_1 - \varpi_0)^T R (\varpi_1 - \varpi_0) \\ &\quad + \left[1 + \frac{1}{\bar{h}} + \frac{3}{\bar{h}-1} + \frac{3}{\bar{h}(\bar{h}-1)} \right] (\varpi_2 - \varpi_1)^T R (\varpi_2 - \varpi_1) \end{aligned} \quad (14)$$

where $\eta(k) := x(k+1) - x(k)$, $\bar{h} := r_2 - r_1$, and

$$\begin{aligned} \varpi_0 &:= x(r_2) - x(r_1) \\ \varpi_1 &:= x(r_1) + x(r_2) - \frac{2}{\bar{h}+1} \sum_{j=r_1}^{r_2} x(j) \\ \varpi_2 &:= \varpi_0 + \frac{6}{(\bar{h}+1)(\bar{h}+2)} \sum_{j=r_1}^{r_2} (r_2+r_1-2j)x(j). \end{aligned}$$

Proof: To begin with, we consider two special cases: $\bar{h} = 1$ and $\bar{h} = 2$. If $\bar{h} = 1$, it is easy to verify that $\varpi_1 = \varpi_2 = 0$, which means that both inequalities (13) and (14) hold. If $\bar{h} = 2$, then $\varpi_2 = 0$. Thus, both inequalities (13) and (14) also hold for $\bar{h} = 2$. In the following, there is only a need to prove the inequalities (13) and (14) for $\bar{h} > 2$.

Proof of (13): Introduce two sequences $\{f_i(j), j \in \mathbb{Z}\}$ ($i = 1, 2$) as

$$\begin{aligned} f_1(j) &:= r_2 + r_1 - 1 - 2j \\ f_2(j) &:= f_1^2(j) - \frac{1}{3}(\bar{h} - 1)(\bar{h} + 1). \end{aligned} \quad (15)$$

Denote $\bar{h} = r_2 - r_1$. Then

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} f_1(j) &= \sum_{j=r_1}^{r_2-1} f_2(j) = \sum_{j=r_1}^{r_2-1} f_1(j)f_2(j) = 0 \\ \sum_{j=r_1}^{r_2-1} f_1^2(j) &= \frac{\bar{h}(\bar{h} - 1)(\bar{h} + 1)}{3} \\ \sum_{j=r_1}^{r_2-1} f_2^2(j) &= \frac{4\bar{h}(\bar{h}^2 - 1)(\bar{h}^2 - 4)}{45} \\ \sum_{j=r_1}^{r_2-1} f_1(j)\eta(j) &= -(\bar{h} + 1)\varpi_1 \\ \sum_{j=r_1}^{r_2-1} f_2(j)\eta(j) &= \frac{2(\bar{h} + 1)(\bar{h} + 2)}{3}\varpi_2. \end{aligned}$$

Let $\varphi(j) := \eta(j) + c_1 f_1(j)\varpi_1 + c_2 f_2(j)\varpi_2$, where c_1 and c_2 are two scalars to be determined. Then

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \varphi^T(j)R\varphi(j) &\geq \frac{1}{\bar{h}} \sum_{j=r_1}^{r_2-1} \varphi^T(j)R \sum_{j=r_1}^{r_2-1} \varphi(j) \\ &= \frac{1}{\bar{h}} \varpi_0^T R \varpi_0. \end{aligned} \quad (16)$$

On the other hand

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \varphi^T(j)R\varphi(j) &= \sum_{j=r_1}^{r_2-1} \left[\eta^T(j)R\eta(j) + c_1^2 f_1^2(j)\varpi_1^T R \varpi_1 \right. \\ &\quad + c_2^2 f_2^2(j)\varpi_2^T R \varpi_2 \\ &\quad + 2c_1 c_2 f_1(j)f_2(j)\varpi_1^T R \varpi_2 \\ &\quad + 2c_1 \varpi_1^T R f_1(j)\eta(j) \\ &\quad \left. + 2c_2 \varpi_2^T R f_2(j)\eta(j) \right]. \end{aligned} \quad (17)$$

After some simple algebraic manipulations, one has

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \varphi^T(j)R\varphi(j) &= \sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) \\ &\quad + \frac{4(\bar{h}+1)(\bar{h}+2)}{45\bar{h}(\bar{h}-1)(\bar{h}-2)} \left\{ \left[c_2 \bar{h}(\bar{h}-1)(\bar{h}-2) + \frac{15}{2} \right]^2 - \frac{225}{4} \right\} \varpi_2^T R \varpi_2 \\ &\quad + \frac{\bar{h}+1}{3\bar{h}(\bar{h}-1)} \left[(c_1 \bar{h}(\bar{h}-1) - 3)^2 - 9 \right] \varpi_1^T R \varpi_1. \end{aligned} \quad (18)$$

Taking $c_1 = [3/(\bar{h}(\bar{h}-1))]$ and $c_2 = -[15/(2\bar{h}(\bar{h}-1)(\bar{h}-2))]$, it follows from (16) and (18) that:

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) &\geq \frac{1}{\bar{h}} \varpi_0^T R \varpi_0 + \frac{3(\bar{h}+1)}{\bar{h}(\bar{h}-1)} \varpi_1^T R \varpi_1 \\ &\quad + \frac{5(\bar{h}+1)(\bar{h}+2)}{\bar{h}(\bar{h}-1)(\bar{h}-2)} \varpi_2^T R \varpi_2 \end{aligned}$$

which gives (13).

Proof of (14): Introduce a sequence $\{f_3(j) : j \in \mathbb{Z}\}$ as

$$f_3(j) := r_2 + 2r_1 - 1 - 3j. \quad (19)$$

Let $\varphi_1(j) = \eta(j) + c_3(\varpi_1 - \varpi_0) + c_4 f_3(j)(\varpi_2 - \varpi_1)$, where c_3 and c_4 are two scalars to be determined. Notice that

$$\sum_{j=r_1}^{r_2-1} (r_2 - j)\varphi_1^T(j)R\varphi_1(j) \geq 0. \quad (20)$$

Expanding the left-hand side of (20), one has

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} (r_2 - j)\eta^T(j)R\eta(j) &+ \frac{\bar{h}+1}{2\bar{h}} \left[(c_3 \bar{h} - 1)^2 - 1 \right] \tilde{\varpi}_1^T R \tilde{\varpi}_1 \\ &+ \frac{(\bar{h}+1)(\bar{h}+2)}{4\bar{h}(\bar{h}-1)} \left\{ [c_4 \bar{h}(\bar{h}-1) + 2]^2 - 4 \right\} \tilde{\varpi}_2^T R \tilde{\varpi}_2 \geq 0 \end{aligned} \quad (21)$$

where $\tilde{\varpi}_1 := \varpi_1 - \varpi_0$ and $\tilde{\varpi}_2 := \varpi_2 - \varpi_1$. Taking $c_3 = (1/\bar{h})$ and $c_4 = -[2/(\bar{h}(\bar{h}-1))]$ yields (14). The proof is completed. ■

Remark 1: In [31] and [32], two finite-sum inequalities are derived for the term $\sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j)$, which are given as

$$\sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) \geq \frac{1}{\bar{h}} \left[\varpi_0^T R \varpi_0 + \frac{3(\bar{h}+1)}{\bar{h}-1} \varpi_1^T R \varpi_1 \right] \quad (22)$$

$$\sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) \geq \frac{1}{\bar{h}} \left[\varpi_0^T R \varpi_0 + \frac{3(\bar{h}-1)}{\bar{h}+1} \varphi_1^T R \varphi_1 \right] \quad (23)$$

where $\varphi_1 := x(r_2) + x(r_1) - [2/(\bar{h}-1)] \sum_{j=r_1+1}^{r_2-1} x(j)$. First, both inequalities (22) and (23) are equivalent because $\varphi_1 = [(\bar{h}+1)/(\bar{h}-1)]\varpi_1$. Second, it is clear to see that the inequality (13) provides a *larger* lower bound than that in (22) as well as that in (23) due to $[(5(\bar{h}+1)(\bar{h}+2))/(\bar{h}(\bar{h}-1)(\bar{h}-2))]\varpi_2^T R \varpi_2 \geq 0$.

Remark 2: From (13), one can see that

$$\sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) \geq \frac{1}{\bar{h}} (\varpi_0^T R \varpi_0 + 3\varpi_1^T R \varpi_1 + 5\varpi_2^T R \varpi_2) \quad (24)$$

$$\begin{aligned} \sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j) &\geq \frac{1}{\bar{h}} \varpi_0^T R \varpi_0 + \left[\frac{3}{\bar{h}} + \frac{6}{\bar{h}(\bar{h}-1)} \right] \varpi_1^T R \varpi_1 \\ &\quad + \left[\frac{5}{\bar{h}} + \frac{10}{\bar{h}(\bar{h}-1)} \right] \varpi_2^T R \varpi_2. \end{aligned} \quad (25)$$

The inequality (24) can be regarded as the discrete-time counterpart of the corresponding integral inequality in [39] and [40]. The inequality (25) offers a tighter bound than that in (24) for the finite-sum term $\sum_{j=r_1}^{r_2-1} \eta^T(j)R\eta(j)$. However, two terms $[6/(\bar{h}(\bar{h}-1))]\varpi_1^T R \varpi_1$ and $[10/(\bar{h}(\bar{h}-1))]\varpi_2^T R \varpi_2$ are not easily dealt with in the estimation on the forward difference of a certain Lyapunov functional. How to handle these useful terms is one of motivations of this paper.

Remark 3: Notice that

$$\sum_{j=r_1}^{r_2-1} (r_2 - j)\eta^T(j)R\eta(j) = \sum_{j=r_1}^{r_2-1} \sum_{i=r_1}^j \eta^T(i)R\eta(i).$$

Thus, the inequality (14) presents a novel bound for the double-sum term, which can be regarded as the discrete-time counterpart of the corresponding integral inequality in [39].

This class of double-sum terms are also used in the H_∞ analysis for discrete-time T-S fuzzy systems with time-varying delays (see [25]).

IV. ROBUST H_∞ PERFORMANCE ANALYSIS

This section focuses on the robust H_∞ performance analysis for the uncertain system (11) using the matrix-based quadratic convex approach together with the novel finite-sum inequalities (24) and (25). In doing so, an augmented FBD Lyapunov functional candidate is chosen as

$$V(k) = V_1(k) + V_2(k) + V_3(k) + V_4(k) + V_5(k) + V_6(k) \quad (26)$$

where

$$\begin{aligned} V_1(k) &:= \varsigma^T(k) \mathcal{P}(k) \varsigma(k) \\ V_2(k) &:= \sum_{j=k-d_1}^{k-1} \vartheta^T(j) \mathcal{Q}_1(j) \vartheta(j) + \sum_{j=k-d_2}^{k-d_1-1} \vartheta^T(j) \mathcal{Q}_2(j) \vartheta(j) \\ V_3(k) &:= d_1 \sum_{j=-d_1}^{-1} \sum_{i=k+j}^{k-1} \eta^T(i) R_1 \eta(i) \\ V_4(k) &:= d_{12} \sum_{j=-d_2}^{-d_1-1} \sum_{i=k+j}^{k-d_1-1} \eta^T(i) R_2 \eta(i) \\ V_5(k) &:= \sum_{j=-d_1}^{-1} \sum_{i=k+j}^{k-1} (-j) \eta^T(i) Z_1 \eta(i) \\ V_6(k) &:= \sum_{j=-d_2}^{-d_1-1} \sum_{i=k+j}^{k-d_1-1} (-d_1-j) \eta^T(i) Z_2 \eta(i) \end{aligned}$$

with $d_{12} := d_2 - d_1$, $\vartheta(k) = \text{col}\{x(k), \eta(k)\}$ and

$$\begin{aligned} \varsigma(k) &:= \text{col}\left\{x(k), x(k-d_1), x(k-d_2), \sum_{j=k-d_1}^{k-1} x(j), \sum_{j=k-d_2}^{k-d_1-1} x(j), \sum_{j=-d_1}^{-1} \sum_{i=k+j}^{k-1} x(i)\right\} \\ \begin{bmatrix} \mathcal{P}(k) \\ \mathcal{Q}_j(k) \end{bmatrix} &= \sum_{i=1}^r h_i(\theta(k)) \begin{bmatrix} P_i \\ Q_{ji} \end{bmatrix}, \quad (j = 1, 2). \end{aligned}$$

Remark 4: Compared with existing FBD Lyapunov functionals, the Lyapunov functional $V(k)$ in (26) has two different characteristics.

- 1) An augmented term in $V_1(k)$ is introduced, which is motivated by the fact that, for nonfuzzy linear systems, an augmented Lyapunov functional usually leads to less conservative results.
- 2) Some new terms in $V_5(k)$ and $V_6(k)$ are given, which can be regarded as triple-sum terms.

Now we state and establish the following result.

Proposition 1: For given $\gamma > 0$ and two positive integers d_1 and d_2 , the closed-loop system described by (11) subject to (8) and (9) is robustly asymptotically stable with a prescribed H_∞ level $\gamma > 0$ if there exist real matrices $P_i > 0, Q_{1i} > 0, Q_{2i} > 0, R_1 > 0, R_2 > 0, Z_1 > 0, Z_2 > 0, U_1 \geq 0, U_2 \geq 0, S_1, S_2, S_3, M_1$, and M_2 with appropriate dimensions, and scalars $\varepsilon_i > 0$ such that

$$\begin{bmatrix} \mathcal{R}_1 & S_1 \\ \star & \mathcal{R}_2 \end{bmatrix} \geq 0, \quad \begin{bmatrix} U_1 & S_2 \\ \star & \mathcal{R}_3 \end{bmatrix} \geq 0, \quad \begin{bmatrix} U_2 & S_3 \\ \star & \mathcal{R}_3 \end{bmatrix} \geq 0 \quad (27)$$

$$2U_1 + 2U_2 + \mathcal{C}_0^T (2P_l - P_i - P_j) \mathcal{C}_0 \geq 0, \quad i \leq j \quad (28)$$

$$\Theta_{ijlst}(d_\sigma) + \Theta_{jilst}(d_\sigma) < 0, \quad i \leq j \quad (29)$$

where $i, j, l, s, t \in \mathcal{S}, \sigma \in \{1, 2\}$ and

$$\Theta_{ijlst}(d_\sigma) := \begin{bmatrix} \Xi_{ijlst}(d_\sigma) & M_0 B_{2i} & \Lambda_{ji}^T & M_0 D_{1i} & \varepsilon_i \Lambda_{2i}^T \\ \star & -\gamma^2 I & B_{3i}^T & 0 & 0 \\ \star & \star & -I & D_{2i} & 0 \\ \star & \star & \star & -\varepsilon_i I & 0 \\ \star & \star & \star & \star & -\varepsilon_i I \end{bmatrix} \quad (30)$$

$$\Xi_{ijlst}(\tau(k)) := \Phi_{ilst} + \Omega_2 + \Omega_3 + \Omega_4 + \text{He}\{M_0 \Psi_{ij}\} \quad (31)$$

$$\begin{cases} \mathcal{R}_1 := \text{diag}\{\tilde{R}_2, d_{12}^{-1} \tilde{Z}_2\}, \mathcal{R}_2 := \text{diag}\{\tilde{R}_2 + \tilde{Z}_2, d_{12}^{-1} \tilde{Z}_2\} \\ \mathcal{R}_3 := \text{diag}\{2d_{12} \tilde{R}_2, 12Z_2\}, \hat{R}_2 := \text{diag}\{3R_2, 5R_2\} \\ \tilde{R}_2 := \text{diag}\{R_2, \hat{R}_2\}, \tilde{Z}_2 := \text{diag}\{Z_2, 3Z_2, 5Z_2\} \\ \hat{Z}_2 := \text{diag}\{2Z_2, 4Z_2\}, M_0 := e_1^T M_1 + e_2^T M_2 \\ \Lambda_{1i} := L_i e_1 + L_{di} e_3, \Lambda_{2i} := E_{1i} e_1 + E_{2i} e_3 \\ \mathcal{C}_0 := \text{col}\{0, 0, 0, 0, e_9 - e_{10}, 0\} \end{cases} \quad (32)$$

with e_ρ ($\rho = 1, 2, \dots, 13$) being the i th $n \times 13n$ row block vector of the $13n \times 13n$ identity matrix, and

$$\Phi_{ilst} := \mathcal{D}_1^T \text{diag}\{P_l, -P_i, Q_{1i}, Q_{2s} - Q_{1s}, -Q_{2t}\} \mathcal{D}_1 \quad (33)$$

$$\Omega_2 := e_2^T \left[d_1^2 R_1 + \frac{d_1(d_1+1)}{2} Z_1 \right] e_2 - \mathcal{C}_3^T \tilde{R}_1 \mathcal{C}_3 - \mathcal{C}_4^T \tilde{Z}_1 \mathcal{C}_4 \quad (34)$$

$$\begin{aligned} \Omega_3 &:= e_5^T \left[d_{12}^2 R_2 + 0.5d_{12}(d_{12}+1)Z_2 \right] e_5 - \tilde{\mathcal{C}}_5^T \tilde{R}_2 \tilde{\mathcal{C}}_5 \\ &\quad - \tilde{\mathcal{C}}_6^T \tilde{R}_2 \tilde{\mathcal{C}}_6 - \left(1 + d_{12}^{-1}\right) \left(\tilde{\mathcal{C}}_7^T \tilde{Z}_2 \tilde{\mathcal{C}}_7 + \tilde{\mathcal{C}}_8^T \tilde{Z}_2 \tilde{\mathcal{C}}_8 \right) \end{aligned} \quad (35)$$

$$\begin{aligned} \Omega_4 &:= \text{He}\left\{ \mathcal{D}_2^T S_1 \mathcal{D}_3 + S_2 \mathcal{D}_4 + S_3 \mathcal{D}_5 \right\} \\ &\quad + \ell_1(k) [\ell_1(k) - 1] U_1 + \ell_2(k) [\ell_2(k) - 1] U_2 \end{aligned} \quad (36)$$

$$\Psi_{ij} := (A_i + B_{1i} K_j - I) e_1 + A_{di} e_3 - e_2 \quad (37)$$

where $\mathcal{C}_1$ and $\mathcal{C}_2$ are defined in (38), shown at the top of the next page, and

$$\begin{cases} \mathcal{C}_3 := \text{col}\{e_1 - e_4, e_1 + e_4 - 2e_8, e_1 - e_4 + 6e_{11}\} \\ \mathcal{C}_4 := \text{col}\{e_4 - e_8, e_4 - e_8 - 3e_{11}\} \\ \mathcal{C}_5 := \text{col}\{e_4 + e_3 - 2e_9, e_4 - e_3 + 6e_{12}\} \\ \mathcal{C}_6 := \text{col}\{e_3 + e_6 - 2e_{10}, e_3 - e_6 + 6e_{13}\} \\ \mathcal{C}_7 := e_3 - e_9 - 3e_{12}, \quad \mathcal{C}_8 := e_6 - e_{10} - 3e_{13} \\ \tilde{\mathcal{C}}_5 := \text{col}\{e_4 - e_3, \mathcal{C}_5\}, \quad \tilde{\mathcal{C}}_6 := \text{col}\{e_3 - e_6, \mathcal{C}_6\} \\ \tilde{\mathcal{C}}_7 := \text{col}\{e_3 - e_9, \mathcal{C}_7\}, \quad \tilde{\mathcal{C}}_8 := \text{col}\{e_6 - e_{10}, \mathcal{C}_8\} \\ \mathcal{D}_1 := \text{col}\{\mathcal{C}_1, \mathcal{C}_2, e_1, e_2, e_4, e_5, e_6, e_7\} \\ \mathcal{D}_2 := \text{col}\{\tilde{\mathcal{C}}_5, \tilde{\mathcal{C}}_7\}, \quad \mathcal{D}_3 := \text{col}\{\tilde{\mathcal{C}}_6, \tilde{\mathcal{C}}_8\} \\ \mathcal{D}_4 := \text{col}\{\mathcal{C}_5, \mathcal{C}_7\}, \quad \mathcal{D}_5 := \text{col}\{\mathcal{C}_6, \mathcal{C}_8\} \end{cases} \quad (39)$$

$$\tilde{R}_1 := \begin{cases} \text{diag}\left\{R_1, \frac{3(d_1+1)}{d_1-1} R_1, \frac{5(d_1+1)(d_1+2)}{(d_1-1)(d_1-2)} R_1\right\}, & d_1 > 2 \\ \text{diag}\{R_1, 3R_1, 5R_1\}, & d_1 \leq 2 \end{cases} \quad (40)$$

$$\tilde{Z}_1 := \begin{cases} \text{diag}\left\{\frac{2(d_1+1)}{d_1} Z_1, \frac{4(d_1+1)(d_1+2)}{d_1(d_1-1)} Z_1\right\}, & d_1 > 1 \\ \text{diag}\{2Z_1, 4Z_1\}, & d_1 = 1 \end{cases} \quad (41)$$

$$\ell_1(k) := \tau(k) - d_1, \quad \ell_2(k) := d_2 - \tau(k). \quad (42)$$

$$\begin{cases} \mathcal{C}_1 := \text{col}\{e_1 + e_2, e_4 + e_5, e_6 + e_7, (d_1 + 1)e_8 - e_4, (\ell_1(k) + 1)e_9 + (\ell_2(k) + 1)e_{10} - e_6 - e_3, \frac{d_1 + 1}{2}[d_1 e_8 - (d_1 + 2)e_{11}]\} \\ \mathcal{C}_2 := \text{col}\{e_1, e_4, e_6, (d_1 + 1)e_8 - e_1, (\ell_1(k) + 1)e_9 + (\ell_2(k) + 1)e_{10} - e_3 - e_4, \frac{(d_1 + 1)(d_1 + 2)}{2}(e_8 - e_{11}) - (d_1 + 1)e_1\} \end{cases} \quad (38)$$

Proof: We first introduce some notations

$$\begin{aligned} \phi_1(k) &:= \frac{1}{d_1 + 1} \sum_{j=k-d_1}^k x(j) \\ \phi_2(k) &:= \frac{1}{\ell_1(k) + 1} \sum_{j=k-\tau(k)}^{k-d_1} x(j) \\ \phi_3(k) &:= \frac{1}{\ell_2(k) + 1} \sum_{j=k-d_2}^{k-\tau(k)} x(j) \\ \phi_4(k) &:= \frac{1}{(d_1 + 1)(d_1 + 2)} \sum_{j=k-d_1}^k (2k - d_1 - 2j)x(j) \\ \phi_5(k) &:= \frac{1}{[\ell_1(k) + 1][\ell_1(k) + 2]} \sum_{j=k-\tau(k)}^{k-d_1} [2k - d_1 - \tau(k) - 2j]x(j) \\ \phi_6(k) &:= \frac{1}{[\ell_2(k) + 1][\ell_2(k) + 2]} \sum_{j=k-d_2}^{k-\tau(k)} [2k - \tau(k) - d_2 - 2j]x(j) \\ \xi(k) &:= \text{col}\{x(k), \eta(k), x(k - \tau(k)), x(k - d_1), \eta(k - d_1) \\ &\quad x(k - d_2), \eta(k - d_2), \phi_1(k) \\ &\quad \phi_2(k), \phi_3(k), \phi_4(k), \phi_5(k), \phi_6(k)\}. \end{aligned}$$

Then taking the forward difference of the Lyapunov functional $V(k)$ yields

$$\Delta V(k) = \sum_{j=1}^6 \Delta V_j(k) \quad (43)$$

where

$$\begin{aligned} \Delta V_1(k) &= \varsigma^T(k + 1)\mathcal{P}(k + 1)\varsigma(k + 1) - \varsigma^T(k)\mathcal{P}(k)\varsigma(k) \\ \Delta V_2(k) &= \vartheta^T(k)\mathcal{Q}_1(k)\vartheta(k) - \vartheta^T(k - d_2)\mathcal{Q}_2(k - d_2)\vartheta(k - d_2) \\ &\quad + \vartheta^T(k - d_1)[\mathcal{Q}_2(k - d_1) - \mathcal{Q}_1(k - d_1)]\vartheta(k - d_1) \\ \Delta V_3(k) &= d_1^2 \eta^T(k)R_1\eta(k) - d_1 \sum_{j=k-d_1}^{k-1} \eta^T(j)R_1\eta(j) \\ \Delta V_4(k) &= d_{12}^2 \eta^T(k - d_1)R_2\eta(k - d_1) - d_{12} \sum_{j=k-d_2}^{k-d_1-1} \eta^T(j)R_2\eta(j) \\ \Delta V_5(k) &= \frac{d_1(d_1 + 1)}{2} \eta^T(k)Z_1\eta(k) - \sum_{j=k-d_1}^{k-1} (k - j)\eta^T(j)Z_1\eta(j) \\ \Delta V_6(k) &= \frac{d_{12}(d_{12} + 1)}{2} \eta^T(k - d_1)Z_2\eta(k - d_1) \\ &\quad - \sum_{j=k-d_2}^{k-d_1-1} (k - d_1 - j)\eta^T(j)Z_2\eta(j). \end{aligned}$$

Note that

$$\varsigma(k + 1) = \mathcal{C}_1 \xi(k), \quad \varsigma(k) = \mathcal{C}_2 \xi(k) \quad (44)$$

where $\mathcal{C}_1$ and $\mathcal{C}_2$ are defined in (38) as shown at the top this page. Thus, one has

$$\Delta V_1(k) + \Delta V_2(k) \leq \xi^T(k)\Omega_1\xi(k) \quad (45)$$

where $\Omega_1 := \mathcal{D}_1^T \Phi(k)\mathcal{D}_1$ with $\mathcal{D}_1$ being defined in (39) and

$$\begin{aligned} \Phi(k) &:= \text{diag}\{\mathcal{P}(k + 1), -\mathcal{P}(k), \mathcal{Q}_1(k) \\ &\quad \mathcal{Q}_2(k - d_1) - \mathcal{Q}_1(k - d_1), -\mathcal{Q}_2(k - d_2)\}. \end{aligned}$$

Applying the inequalities (13) and (14) yields, respectively

$$-d_1 \sum_{j=k-d_1}^{k-1} \eta^T(j)R_1\eta(j) \leq -\xi^T(k)\mathcal{C}_3^T \tilde{R}_1 \mathcal{C}_3 \xi(k) \quad (46)$$

$$- \sum_{j=k-d_1}^{k-1} (k - j)\eta^T(j)Z_1\eta(j) \leq -\xi^T(k)\mathcal{C}_4^T \tilde{Z}_1 \mathcal{C}_4 \xi(k) \quad (47)$$

where $\mathcal{C}_3$ and $\mathcal{C}_4$, $\tilde{R}_1$, and $\tilde{Z}_1$ are defined in (39)–(41), respectively. In consequence

$$\Delta V_3(k) + \Delta V_5(k) \leq \xi^T(k)\Omega_2\xi(k) \quad (48)$$

where Ω_2 is defined in (34). For the finite-sum terms in $\Delta V_4(k)$ and $\Delta V_6(k)$, denote

$$\alpha_1(k) := -d_{12} \sum_{j=k-d_2}^{k-d_1-1} \eta^T(j)R_2\eta(j) \quad (49)$$

$$\alpha_2(k) := - \sum_{j=k-d_2}^{k-d_1-1} (k - d_1 - j)\eta^T(j)Z_2\eta(j). \quad (50)$$

It is clear to see that

$$\alpha_1(k) = -d_{12} \sum_{j=k-d_2}^{k-\tau(k)-1} \eta^T(j)R_2\eta(j) - d_{12} \sum_{j=k-\tau(k)}^{k-d_1-1} \eta^T(j)R_2\eta(j).$$

Applying the inequality (25), one has

$$\begin{aligned} \alpha_1(k) &\leq -\xi^T(k) \left\{ \frac{d_{12}}{\ell_1(k)} \tilde{\mathcal{C}}_5^T \tilde{R}_2 \tilde{\mathcal{C}}_5 + \frac{d_{12}}{\ell_2(k)} \tilde{\mathcal{C}}_6^T \tilde{R}_2 \tilde{\mathcal{C}}_6 \right\} \xi(k) \\ &\quad - \frac{2d_{12}}{\ell_1(k)[\ell_1(k) - 1]} \xi^T(k) \tilde{\mathcal{C}}_5^T \tilde{R}_2 \tilde{\mathcal{C}}_5 \xi(k) \\ &\quad - \frac{2d_{12}}{\ell_2(k)[\ell_2(k) - 1]} \xi^T(k) \tilde{\mathcal{C}}_6^T \tilde{R}_2 \tilde{\mathcal{C}}_6 \xi(k) \end{aligned} \quad (51)$$

where $\mathcal{C}_5$ and $\mathcal{C}_6$ are defined in (39). Observe that

$$\begin{aligned} \alpha_2(k) &= - \sum_{j=k-d_2}^{k-\tau(k)-1} (k - \tau(k) - j)\eta^T(j)Z_2\eta(j) \\ &\quad - \sum_{j=k-\tau(k)}^{k-d_1-1} (k - d_1 - j)\eta^T(j)Z_2\eta(j) \\ &\quad - \ell_1(k) \sum_{j=k-d_2}^{k-\tau(k)-1} \eta^T(j)Z_2\eta(j). \end{aligned} \quad (52)$$

Applying the inequalities (14) and (24) yields

$$\begin{aligned} \alpha_2(k) &\leq -\frac{\ell_1(k)}{\ell_2(k)} \xi^T(k) \tilde{\mathcal{C}}_6^T \tilde{Z}_2 \tilde{\mathcal{C}}_6 \xi(k) \\ &\quad - \left[1 + \frac{1}{\ell_1(k)} \right] \xi^T(k) \tilde{\mathcal{C}}_7^T \tilde{Z}_2 \tilde{\mathcal{C}}_7 \xi(k) \\ &\quad - \left[1 + \frac{1}{\ell_2(k)} \right] \xi^T(k) \tilde{\mathcal{C}}_8^T \tilde{Z}_2 \tilde{\mathcal{C}}_8 \xi(k) \\ &\quad - \frac{12}{\ell_1(k)[\ell_1(k) - 1]} \xi^T(k) \tilde{\mathcal{C}}_7^T \tilde{Z}_2 \tilde{\mathcal{C}}_7 \xi(k) \\ &\quad - \frac{12}{\ell_2(k)[\ell_2(k) - 1]} \xi^T(k) \tilde{\mathcal{C}}_8^T \tilde{Z}_2 \tilde{\mathcal{C}}_8 \xi(k) \end{aligned} \quad (53)$$

where $\mathcal{C}_7$, $\tilde{\mathcal{C}}_7$, $\mathcal{C}_8$, and $\tilde{\mathcal{C}}_8$ are defined in (39), and $\tilde{Z}_2$ and $\hat{Z}_2$ in (32). From (51) and (53), one has

$$\Delta V_4(k) + \Delta V_6(k) \leq \xi^T(k)\Omega_3\xi(k) + \sum_{i=1}^3 \beta_i(k) \quad (54)$$

where Ω_3 is defined in (35), and

$$\beta_1(k) := -\xi^T(k) \left[\frac{\ell_2(k)}{\ell_1(k)} \mathcal{D}_2^T \mathcal{R}_1 \mathcal{D}_2 + \frac{\ell_1(k)}{\ell_2(k)} \mathcal{D}_3^T \mathcal{R}_2 \mathcal{D}_3 \right] \xi(k)$$

$$\beta_2(k) := -\frac{1}{\ell_1(k)[\ell_1(k)-1]} \xi^T(k) \mathcal{D}_4^T \mathcal{R}_3 \mathcal{D}_4 \xi(k) \quad (55)$$

$$\beta_3(k) := -\frac{1}{\ell_2(k)[\ell_2(k)-1]} \xi^T(k) \mathcal{D}_5^T \mathcal{R}_3 \mathcal{D}_5 \xi(k) \quad (56)$$

where $\mathcal{D}_2, \dots, \mathcal{D}_5$ are defined in (39), $\mathcal{R}_1, \mathcal{R}_2$, and $\mathcal{R}_3$ in (32). Employing the reciprocally convex approach yields

$$\beta_1(k) \leq 2\xi^T(k) \mathcal{D}_2^T S_1 \mathcal{D}_3 \xi(k).$$

Apply Lemma 2 to get

$$\beta_2(k) \leq \ell_1(k)[\ell_1(k)-1] \xi^T(k) U_1 \xi(k) + 2\xi^T(k) S_2 \mathcal{D}_4 \xi(k)$$

$$\beta_3(k) \leq \ell_2(k)[\ell_2(k)-1] \xi^T(k) U_2 \xi(k) + 2\xi^T(k) S_3 \mathcal{D}_5 \xi(k)$$

which lead to

$$\Delta V_4(k) + \Delta V_6(k) \leq \xi^T(k) [\Omega_3 + \Omega_4] \xi(k) \quad (57)$$

where Ω_4 is defined in (36). To summarize, the forward difference $\Delta V(k)$ satisfies

$$\Delta V(k) \leq \xi^T(k) (\Omega_1 + \Omega_2 + \Omega_3 + \Omega_4) \xi(k). \quad (58)$$

In the following, we complete the proof from two aspects: 1) the system (12) is robust asymptotically stable and 2) under zero conditions, for any nonzero $w(k) \in l_2$, the inequality $\|z\|_2 < \gamma \|w\|_2$ holds if the matrix inequalities in (27) and (29) are satisfied.

- 1) It is proven that the system (12) is robust asymptotically stable if the matrix inequalities in (27)–(29) hold. In fact, for any real matrices M_1 and M_2 of compatible dimensions, one has

$$0 = 2\xi^T(k) (e_1^T M_1 + e_2^T M_2) \times [(\mathcal{A}(k) - I)e_1 + \mathcal{A}_d(k)e_3 - e_2] \xi(k)$$

which, together with (58), gives

$$\begin{aligned} \Delta V(k) &\leq \sum_{i=1}^r \sum_{j=1}^r \sum_{l=1}^r \sum_{s=1}^r \sum_{t=1}^r \xi^T(k) h_{ij} h_{lst} \Upsilon_{ijlst}(\tau(k)) \xi(k) \\ &= \sum_{l=1}^r \sum_{s=1}^r \sum_{t=1}^r h_{lst} \sum_{i=1}^r \sum_{j=1}^r \xi^T(k) h_{ij} \Upsilon_{ijlst}(\tau(k)) \xi(k) \\ &= \sum_{l=1}^r \sum_{s=1}^r \sum_{t=1}^r h_{lst} \left(\sum_{i=1}^r \xi^T(k) h_{ii} \Upsilon_{iilst}(\tau(k)) \xi(k) \right. \\ &\quad \left. + \sum_{i=1}^{r-1} \sum_{j=i+1}^r \xi^T(k) h_{ij} \Gamma_{ijlst}(\tau(k)) \xi(k) \right) \end{aligned} \quad (59)$$

where

$$h_{ij} := h_i(\theta(k)) h_j(\theta(k))$$

$$h_{lst} := h_l(\theta(k+1)) h_s(\theta(k-d_1)) h_t(\theta(k-d_2))$$

$$\Gamma_{ijlst}(\tau(k)) := \Upsilon_{ijlst}(\tau(k)) + \Upsilon_{jilst}(\tau(k))$$

$$\begin{aligned} \Upsilon_{ijlst}(\tau(k)) &:= \Phi_{ilst} + \Omega_2 + \Omega_3 + \Omega_4 \\ &\quad + \text{He} \{ M_0 [\Psi_{ij} + D_{1i} F_i(k) \Lambda_{2i}] \} \end{aligned} \quad (60)$$

with Φ_{ilst}, M_0 and Ψ_{ij}, Λ_{2i} being defined in (33) and (32), respectively.

Since $h_i(\cdot) \geq 0$ and $\sum_{i=1}^r h_i(\cdot) = 1$, it follows from (59) that the inequalities $\Upsilon_{ijlst}(\tau(k)) < 0, i, j, l, s, t \in \mathcal{S}$ hold if the following inequalities are satisfied:

$$\Upsilon_{iilst}(\tau(k)) < 0, i, l, s, t \in \mathcal{S} \quad (61)$$

$$\Gamma_{ijlst}(\tau(k)) < 0, i, j, l, s, t \in \mathcal{S}; i < j \quad (62)$$

which can be rewritten as

$$\Gamma_{ijlt}(\tau(k)) < 0, i, j, l, s, t \in \mathcal{S}; i \leq j. \quad (63)$$

If $\Upsilon_{ijlst}(\tau(k)) < 0, i, j, l, s, t \in \mathcal{S}$, then there exists a scalar $v > 0$ such that $\Delta V(k) \leq -v \xi^T(k) \xi(k) \leq -v x^T(k) x(k) < 0$ for $x(k) \neq 0$, which means that the system (12) is robustly asymptotically stable. In the following, one can see that (63) is implied by (29). First, noting that $\Gamma_{ijlst}(\tau(k))$ is a quadratic matrix-valued function on $\tau(k)$, and the coefficient of $\tau^2(k)$ is given by

$$\Gamma_{0ijl} := 2U_1 + 2U_2 + \mathcal{C}_0^T (2P_l - P_i - P_j) \mathcal{C}_0, l \in \mathcal{S}; i \leq j$$

with $\mathcal{C}_0$ given in (32). By Lemma 1, $\Gamma_{ijlst}(\tau(k)) < 0$ is ensured if $\Gamma_{0ijl} \geq 0$ and $\Gamma_{ijlst}(d_1) < 0$ and $\Gamma_{ijlst}(d_2) < 0$ hold. Second, for $F_i(k)$ satisfying (8), one has $\Upsilon_{ijlst}(d_\sigma) < 0$ for $i, j, l, s, t \in \mathcal{S}; \sigma \in \{1, 2\}$ if

$$\Gamma_{ijlst}(d_\sigma) = \Upsilon_{ijlst}(d_\sigma) + \Upsilon_{jilst}(d_\sigma) < 0, i \leq j \quad (64)$$

where

$$\Upsilon_{ijlst}(d_\sigma) = \begin{bmatrix} \Xi_{ijlst}(d_\sigma) & M_0 D_{1i} & \varepsilon_i \Lambda_{2i}^T \\ \star & -\varepsilon_i I & 0 \\ \star & \star & -\varepsilon_i I \end{bmatrix}.$$

Obviously, (63) is implied by (29).

- 2) It is proven that under zero initial conditions the inequality $\|z\|_2 < \gamma \|w\|_2$ is satisfied for any nonzero $w(k) \in l_2$. Suppose that $w(k) \neq 0$. Then, for any real matrices M_1 and M_2 of compatible dimensions, one has

$$0 = 2\xi^T(k) (e_1^T M_1 + e_2^T M_2) \times \{[(\mathcal{A}(k) - I)e_1 + \mathcal{A}_d(k)e_3 - e_2] \xi(k) + \mathcal{B}_1(k)w(k)\}.$$

Hence

$$\begin{aligned} \Delta V(k) + z^T(k) z(k) - \gamma^2 w^T(k) w(k) \\ \leq \sum_{i=1}^r \sum_{j=1}^r \sum_{l=1}^r \sum_{s=1}^r \sum_{t=1}^r h_{ij} h_{lst} \zeta^T(k) \hat{\Upsilon}_{ijlst}(\tau(k)) \zeta(k) \end{aligned}$$

where

$$\zeta(k) := \text{col}\{\xi(k), w(k)\}$$

$$\begin{aligned} \hat{\Upsilon}_{ijlst}(\tau(k)) &:= \begin{bmatrix} \Upsilon_{ijlst}(\tau(k)) & M_0 B_{2i} \\ \star & -\gamma^2 I \end{bmatrix} \\ &\quad + \begin{bmatrix} \hat{L}_i^T \\ B_{3i}^T \end{bmatrix} \begin{bmatrix} \hat{L}_i & B_{3i} \end{bmatrix} \end{aligned}$$

$$\hat{L}_i := \Lambda_{1i} + D_{2i} F_i(k) \Lambda_{2i}.$$

Similar to (59) and (61)–(63), it is not difficult to prove that $\Delta V(k) + z^T(k) z(k) - \gamma^2 w^T(k) w(k) < 0$ if

$$\hat{\Upsilon}_{ijlst}(d_\sigma) + \hat{\Upsilon}_{jilst}(d_\sigma) < 0, i \leq j \quad (65)$$

for $i, j, l, s, t \in \mathcal{S}; \sigma \in \{1, 2\}$, where

$$\hat{\Upsilon}_{ijlst}(d_\sigma) = \begin{bmatrix} \Upsilon_{ijlst}(d_\sigma) & M_0 B_{2i} & \hat{L}_i^T \\ \star & -\gamma^2 I & B_{3i}^T \\ \star & \star & -I \end{bmatrix}$$

which is equivalent to

$$\Theta_{ijlst}(d_\sigma) + \Theta_{jilst}(d_\sigma) < 0, \quad i \leq j \quad (66)$$

where $\Theta_{ijlst}(d_\sigma)$ is given in (30). By applying matrix-based quadratic convex approach, (66) holds if the matrix inequalities (27)–(29) are satisfied. In this situation, one has

$$\sum_{k=0}^{\infty} [\Delta V(k) + z^T(k)z(k) - \gamma^2 w^T(k)w(k)] < 0. \quad (67)$$

Under zero conditions, it is clear to see that

$$\sum_{k=0}^{\infty} [z^T(k)z(k) - \gamma^2 w^T(k)w(k)] < 0$$

which leads to $\|z\|_2 < \gamma \|w\|_2$. The proof is thus completed. ■

Remark 5: Proposition 1 presents a new BRL for the closed-loop system (11). Compared with some existing results reported in the literature, Proposition 1 is of less conservatism due to the following facts.

- 1) Two new finite-sum inequalities (13) and (14) are used to provide a tighter estimation on the forward difference of $V(k)$ than that using the Jensen-type inequality, the Wirtinger-based finite-sum inequality [32] and the Abel lemma-based finite-sum inequality [31].
- 2) When the finite-sum inequalities (13) and (14) are applied, some terms, i.e., $\beta_2(k)$ and $\beta_3(k)$ in (55) and (56), respectively, are not overly bounded by zero. In fact, it is not easy to deal with these terms since quadratic polynomials as $\ell_1(k)[\ell_1(k) - 1]$ and $\ell_2(k)[\ell_2(k) - 1]$ appear in the forms of their inverse. Nevertheless, in this paper, $\beta_2(k)$ and $\beta_3(k)$ are bounded using Lemma 2. As a result, those quadratic polynomials appear in the forward difference $\Delta V(k)$ linearly. Then the idea of a matrix-based quadratic convex approach is employed to formulate a novel BRL for the system (11).
- 3) Another point is that the last term in $\alpha_2(k)$ in (52) is also retained (not overly bounded by zero) in the forward difference $\Delta V(k)$. Consequently, when the reciprocally convex approach is applied, the matrix constraint $\begin{bmatrix} \bar{\mathcal{R}}_1 & \bar{\mathcal{S}}_1 \\ \star & \bar{\mathcal{R}}_1 \end{bmatrix} \geq 0$ can be relaxed as $\begin{bmatrix} \bar{\mathcal{R}}_1 & \bar{\mathcal{S}}_1 \\ \star & \bar{\mathcal{R}}_2 \end{bmatrix} \geq 0$.

However, since the augmented matrix $\bar{\mathcal{P}}(k)$ and some extra matrix variables as U_1, U_2, S_1, S_2 , and S_3 are introduced, Proposition 1 is of higher computation complexity.

Remark 6: Proposition 1 also provides a different method from [37] and [38] for studying stability of linear discrete-time systems with time-varying delays. That is, some novel finite-sum inequalities are first used to bound the finite-sum terms $\alpha_1(k)$ and $\alpha_2(k)$ appearing in $\Delta V(k)$, then employ a matrix-based quadratic convex approach to ensure that $\Delta V(k)$ is upper bounded as tight as possible, e.g., $\beta_1(k)$ and $\beta_2(k)$ not overly bounded by zero. However, in [37], a new Lyapunov

functional with a delay-product type term is introduced while bounding $\alpha_1(k)$ and $\alpha_2(k)$ is similar to the use of the Wirtinger-based finite-sum inequality. Although some general finite-sum inequalities are derived in [38], the terms $\beta_1(k)$ and $\beta_2(k)$ are directly enlarged by zero in the stability analysis.

V. ROBUST H_∞ CONTROL DESIGN

In this section, based on Proposition 1, the robust fuzzy H_∞ control problem is solved in terms of a set of linear matrix inequalities. Note that the matrix inequalities in (28) are nonlinear due to the nonlinear terms $M_1(A_i + B_{1i}K_j)$ and $M_2(A_i + B_{1i}K_j)$. Proposition 2 below provides a simple method to work out the controller gains K_j ($j \in \mathcal{S}$).

Proposition 2: For given $\gamma > 0$ and $\varrho > 0$ and two positive integers d_1 and d_2 , the robust fuzzy H_∞ control problem is solvable with $K_j = Y_j \bar{X}^{-T}$ if there exist real matrices $\bar{P}_i > 0$, $\bar{Q}_{1i} > 0$, $\bar{Q}_{2i} > 0$, $\bar{R}_1 > 0$, $\bar{R}_2 > 0$, $\bar{Z}_1 > 0$, $\bar{Z}_2 > 0$, $\bar{U}_1 \geq 0$, $\bar{U}_2 \geq 0$, $\bar{S}_1$, $\bar{S}_2$, $\bar{S}_3$, $\bar{X}$, and Y_j with appropriate dimensions, and scalars $\tilde{\varepsilon}_i > 0$ such that

$$\begin{bmatrix} \bar{\mathcal{R}}_1 & \bar{\mathcal{S}}_1 \\ \star & \bar{\mathcal{R}}_2 \end{bmatrix} \geq 0, \quad \begin{bmatrix} \bar{U}_1 & \bar{\mathcal{S}}_2 \\ \star & \bar{\mathcal{R}}_3 \end{bmatrix} \geq 0, \quad \begin{bmatrix} \bar{U}_2 & \bar{\mathcal{S}}_3 \\ \star & \bar{\mathcal{R}}_3 \end{bmatrix} \geq 0 \quad (68)$$

$$2\bar{U}_1 + 2\bar{U}_2 + \mathcal{C}_0^T(2\bar{P}_1 - \bar{P}_i - \bar{P}_j)\mathcal{C}_0 \geq 0, \quad i \leq j \quad (69)$$

$$\bar{\Theta}_{ijlst}(d_\sigma) + \bar{\Theta}_{jilst}(d_\sigma) < 0, \quad i \leq j \quad (70)$$

where $i, j, l, s, t \in \mathcal{S}; \sigma \in \{1, 2\}$

$$\bar{\Theta}_{ijlst}(d_\sigma) := \begin{bmatrix} \bar{\Xi}_{ijlst}(d_\sigma) & \mathcal{D}_0 B_{2i} & \bar{\Lambda}_{1i}^T & \tilde{\varepsilon}_i \mathcal{D}_0 D_{1i} & \bar{\Lambda}_{2i}^T \\ \star & -\gamma^2 I & B_{3i}^T & 0 & 0 \\ \star & \star & -I & \tilde{\varepsilon}_i D_{2i} & 0 \\ \star & \star & \star & -\tilde{\varepsilon}_i I & 0 \\ \star & \star & \star & \star & -\tilde{\varepsilon}_i I \end{bmatrix}$$

$$\bar{\Xi}_{ijlst}(\tau(k)) := \bar{\Phi}_{ilst} + \bar{\Omega}_2 + \bar{\Omega}_3 + \bar{\Omega}_4 + \text{He}\{\mathcal{D}_0 \bar{\Psi}_{ij}\}$$

$$\bar{\Lambda}_{1i} := L_i \bar{X}^T e_1 + L_{di} \bar{X}^T e_3$$

$$\bar{\Lambda}_{2i} := E_{1i} \bar{X}^T e_1 + E_{2i} \bar{X}^T e_3, \quad \mathcal{D}_0 := e_1^T + \varrho e_2^T$$

$$\bar{\Psi}_{ij} := [(A_i - I)\bar{X} + B_{1i}Y_j]e_1 + A_{di}\bar{X}e_3 - \bar{X}e_2$$

and $\bar{\mathcal{R}}_1, \bar{\mathcal{R}}_2, \bar{\mathcal{R}}_3, \bar{\Phi}_{ijlst}, \bar{\Omega}_2, \bar{\Omega}_3$, and $\bar{\Omega}_4$ are obtained from $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \Phi_{ijlst}, \Omega_2, \Omega_3$, and Ω_4 by replacing the matrices $P_i, Q_{1i}, Q_{2i}, R_{1i}, R_{2i}, Z_{1i}, Z_{2i}, S_1, S_2, S_3, U_1$, and U_2 with $\bar{P}_i, \bar{Q}_{1i}, \bar{Q}_{2i}, \bar{R}_{1i}, \bar{R}_{2i}, \bar{Z}_{1i}, \bar{Z}_{2i}, \bar{S}_1, \bar{S}_2, \bar{S}_3, \bar{U}_1$, and $\bar{U}_2$, respectively.

Proof: We complete the proof based on Proposition 1. Let $M_1 = X$ and $M_2 = \varrho X$, where X is a nonsingular real matrix and $\varrho > 0$ is a tuning parameter. Denote

$$J_0 := \sum_{j=1}^{13} e_j^T X^{-1} e_j, \quad J := \text{diag}\{J_0, I, I, \varepsilon_i^{-1} I, \varepsilon_i^{-1} I\}.$$

Left- and right-multiplying the left-hand side of (29) by J and J^T , respectively, yields

$$J[\Theta_{ijlst}(d_\sigma) + \Theta_{jilst}(d_\sigma)]J^T < 0, \quad i \leq j. \quad (71)$$

Denote $\tilde{\varepsilon}_i := \varepsilon_i^{-1}$, $\bar{X} := X^{-1}$ and

$$X_1 := \text{diag}\{X^{-1}, X^{-1}\}, \quad X_2 := \text{diag}\{X^{-1}, X^{-1}, X^{-1}\}$$

$$\bar{Q}_{1i} := X_1 Q_{1i} X_1^T, \quad \bar{Q}_{2i} := X_1 Q_{2i} X_1^T$$

$$\bar{R}_{1i} := X^{-1} R_{1i} X^{-T}, \quad \bar{R}_{2i} := X^{-1} R_{2i} X^{-T}$$

$$\bar{Z}_{1i} := X^{-1} Z_{1i} X^{-T}, \quad \bar{Z}_{2i} := X^{-1} Z_{2i} X^{-T}$$

$$\begin{aligned}\bar{P}_i &:= \text{diag}\{X_2, X_2\}P_i\text{diag}\{X_2^T, X_2^T\}, \quad \bar{U}_1 := J_0U_1J_0^T \\ \bar{S}_1 &:= \text{diag}\{X_1, X_2\}S_1\text{diag}\{X_1^T, X_2^T\}, \quad \bar{U}_2 := J_0U_2J_0^T \\ \bar{S}_2 &:= J_0S_2X_2^T, \quad \bar{S}_3 := J_0S_3X_2^T, \quad Y_i := K_iX^{-T}.\end{aligned}$$

Then (71) becomes (70). On the other hand, it is clear that the matrix inequalities in (68) and (69) are equivalent to the corresponding ones in (27) and (28). Thus, by Proposition 1, if the matrix inequalities (68)–(70) are satisfied, the fuzzy controller (10) with $K_i = Y_i\bar{X}^{-T}$ renders the resultant closed-loop system robustly asymptotically stable with a prescribed H_∞ level $\gamma > 0$, which completes the proof. ■

Remark 7: Proposition 2 presents a design method of fuzzy H_∞ controllers, which depends on a tuning parameter $\varrho > 0$. By using a binary search algorithm, an optimal value of ϱ can be obtained for a prescribed H_∞ level $\gamma > 0$. The proposed method is different from those in [25], [26], and [30], where some nonlinear terms such as $-W^TRW$ are replaced with $R - W^T - W$, which usually leads to conservative results. Moreover, the proposed method can be also employed to deal with the dissipative control [43], [44], H_∞ and $l_2 - l_\infty$ filtering [45]–[47] and stochastic and output feedback control [48]–[50] for discrete-time T-S fuzzy systems with interval-like time-varying delay.

Notice that Proposition 2 is suitable for the T-S fuzzy system (7), where both saturation and measurement noise in the control signal are *not* taken into account. As a complement, in the following, we present some result on the H_∞ control for a T-S fuzzy system with both input saturation and measurement noise in the control signal by employing the proposed method. In doing so, consider the uncertain T-S fuzzy model of the form.

Plant Rule i: IF $\theta_1(k)$ is μ_{i1} , $\theta_2(k)$ is $\mu_{i2}, \dots$, and $\theta_s(k)$ is μ_{is} , THEN

$$\begin{cases} x(k+1) = \tilde{A}_i x(k) + \tilde{A}_{di} x(k - \tau(k)) + B_{1i} \tilde{u}(k) + B_{2i} w(k) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2 + 1, \dots, 0\} \\ z(k) = \tilde{L}_i x(k) + \tilde{L}_{di} x(k - \tau(k)) + B_{3i} w(k) \end{cases} \quad (72)$$

where $\tilde{u}(k) = \text{sat}(u(k))$ is the saturated control input and $u(k)$ is the state feedback fuzzy controller of the form.

Control Rule i: IF $\theta_1(k)$ is μ_{i1} , $\theta_2(k)$ is $\mu_{i2}, \dots$, and $\theta_s(k)$ is μ_{is} , THEN

$$u(k) = K_i x(k) + G_i v(k) \quad (73)$$

where $v(k)$ represents an energy bounded measurement noise belonging to l_2 , and K_i, G_i are controller gain matrices to be designed.

The saturation function $\text{sat}(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is defined as

$$\text{sat}(u) = \text{col}\{\text{sat}_1(u_1), \text{sat}_2(u_2), \dots, \text{sat}_m(u_m)\}$$

where $\text{sat}_i(u_i) = \text{sign}(u_i) \min\{u_{i,\max}, |u_i|\}$, for all $i = 1, 2, \dots, m$, with $u_{i,\max}$ denoting the i th element of the vector $u_{\max}$.

Similar to some existing results regarding actuator saturation or sensor saturation (see [41]), it is assumed that there exist

two diagonal matrices H_1 and H_2 such that $0 \leq H_1 < I \leq H_2$. Then the saturation function $\text{sat}(u(k))$ in (72) can be written as

$$\text{sat}(u(k)) = H_1 u(k) + \sigma(u(k))$$

where $\sigma(u(k))$ is a nonlinear vector-valued function satisfying the following sector condition with $K_1 = 0$ and $K_2 = H = H_2 - H_1$ [42]:

$$\sigma^T(u(k))(\sigma(u(k)) - Hu(k)) \leq 0.$$

Correspondingly, the closed-loop system is obtained as

$$\begin{cases} x(k+1) = \bar{A}(k)x(k) + \bar{A}_d(k)x(k - \tau(k)) \\ \quad + \bar{B}_1(k)w(k) + \bar{B}_2(k)v(k) + \bar{C}(k)\sigma(u(k)) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2 + 1, \dots, 0\} \\ z(k) = \bar{\mathcal{L}}(k)x(k) + \bar{\mathcal{L}}_d(k)x(k - \tau(k)) + \bar{B}_2(k)w(k) \end{cases} \quad (74)$$

where

$$\begin{aligned}\bar{A}(k) &= \sum_{i=1}^r \sum_{j=1}^r h_i(\theta(k))h_j(\theta(k))\bar{\mathcal{A}}_{ij}(k) \\ \bar{\mathcal{A}}_{ij}(k) &= A_i + B_{1i}H_1K_j + D_{1i}F_i(k)E_{1i} \\ \bar{B}_1(k) &= \sum_{i=1}^r \sum_{j=1}^r h_i(\theta(k))h_j(\theta(k))B_{1i}H_1G_j \\ \bar{C}(k) &= \sum_{i=1}^r h_i(\theta(k))B_{1i}.\end{aligned}$$

The quadratic H_∞ performance function is defined as

$$\tilde{J}_{w,v} = \|z(k)\|_2 - \gamma \|w(k)\|_2 - \gamma \|v(k)\|_2$$

where $\gamma > 0$ is the prescribed H_∞ level. The objective is thus modified to design suitable fuzzy controllers of the form (73) such that the closed-loop system (74) is robustly asymptotically stable with $\tilde{J}_{w,v} < 0$ under zero initial conditions for any nonzero $w(k), v(k) \in l_2$.

Applying the proposed finite-sum inequality-based method plus a matrix-based quadratic convex approach, the following result is readily derived to solve out the above H_∞ fuzzy control problem for the system (74), which can be used to consider the effects of input saturation and measurement noise on the closed-loop system performance.

Proposition 3: For given $\gamma > 0$ and $\varrho > 0$, and two positive integers d_1 and d_2 , the robust fuzzy H_∞ control problem is solvable with $K_j = Y_j\bar{X}^{-T}$ if there exist real matrices $\bar{P}_i > 0$, $\bar{Q}_{1i} > 0$, $\bar{Q}_{2i} > 0$, $\bar{R}_1 > 0$, $\bar{R}_2 > 0$, $\bar{Z}_1 > 0$, $\bar{Z}_2 > 0$, $\bar{U}_1 \geq 0$, $\bar{U}_2 \geq 0$, $\bar{S}_1, \bar{S}_2, \bar{S}_3, \bar{X}, Y_j$, and G_j with appropriate dimensions, and scalars $\kappa \geq 0$, $\tilde{\epsilon}_i > 0$ such that

$$\begin{bmatrix} \bar{R}_1 & \bar{S}_1 \\ \star & \bar{R}_2 \end{bmatrix} \geq 0, \quad \begin{bmatrix} \bar{U}_1 & \bar{S}_2 \\ \star & \bar{R}_3 \end{bmatrix} \geq 0, \quad \begin{bmatrix} \bar{U}_2 & \bar{S}_3 \\ \star & \bar{R}_3 \end{bmatrix} \geq 0 \quad (75)$$

$$2\bar{U}_1 + 2\bar{U}_2 + \mathcal{C}_0^T(2\bar{P}_l - \bar{P}_i - \bar{P}_j)\mathcal{C}_0 \geq 0, \quad i \leq j \quad (76)$$

$$\tilde{\Theta}_{ijlst}(d_\sigma) + \tilde{\Theta}_{jilst}(d_\sigma) < 0, \quad i \leq j \quad (77)$$

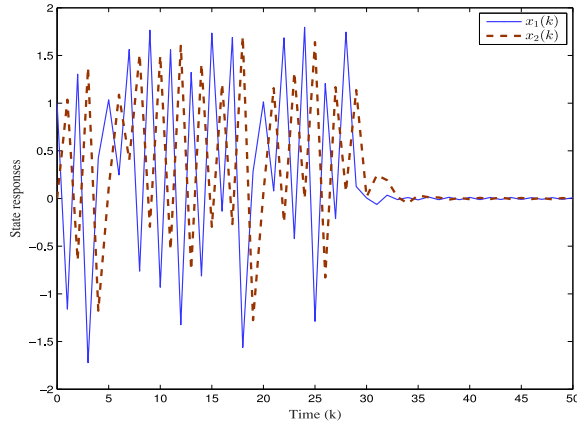


Fig. 1. Initial state responses for Example 1.

for all $i, j, l, s, t \in \mathcal{S}$; $\sigma \in \{1, 2\}$, where

$$\tilde{\Theta}_{ijlst}(d_\sigma) := \begin{bmatrix} \tilde{\Theta}_{ijlst}(d_\sigma) & F_{ij}^{12} & \mathcal{D}_0 B_{2i} & F_{ij}^{14} & \bar{\Lambda}_{1i}^T & F_{ij}^{16} & \bar{\Lambda}_{2i}^T \\ \star & -2\kappa I & 0 & \kappa H G_j & 0 & 0 & 0 \\ \star & \star & -\gamma^2 I & 0 & B_{3i}^T & 0 & 0 \\ \star & \star & \star & -\gamma^2 I & 0 & 0 & 0 \\ \star & \star & \star & \star & -I & \tilde{\varepsilon}_i D_{2i} & 0 \\ \star & \star & \star & \star & \star & -\tilde{\varepsilon}_i I & 0 \\ \star & \star & \star & \star & \star & \star & -\tilde{\varepsilon}_i I \end{bmatrix}$$

with $\tilde{\Theta}_{ijlst}(d_\sigma) := \bar{\Phi}_{ilst} + \bar{\Omega}_2 + \bar{\Omega}_3 + \bar{\Omega}_4 + \text{He}\{\mathcal{D}_0 \tilde{\Psi}_{ij}\}$, $\tilde{\Psi}_{ij} := [(A_i - D)\bar{X} + B_{1i}H_1Y_j]e_1 + A_{di}\bar{X}e_3 - \bar{X}e_2$, $F_{ij}^{12} := \mathcal{D}_0 B_{1i} + e_1^T \kappa Y_j^T H^T$, $F_{ij}^{14} := \mathcal{D}_0 B_{1i}H_1G_j$, $F_{ij}^{16} := \tilde{\varepsilon}_i \mathcal{D}_0 D_{1i}$, and the other notations are defined in Proposition 2.

Proof: The conclusion can be drawn by following the similar proofs of Propositions 1 and 2. It is omitted due to page limitation. ■

VI. ILLUSTRATIVE EXAMPLES

In this section, three examples are provided to illustrate the effectiveness and advantage of the proposed method. The first and second examples are used for the purpose of comparing with some existing results, and the third example is for the purpose of an application of the obtained results.

Example 1: Consider the following Hénon-like system [26]:

$$\begin{cases} x_1(k+1) = -\{cx_1(k) + (1-c)x_1(k-\tau(k))\}^2 \\ \quad + 0.3x_2(k) + 1.4 + u(k) \\ x_2(k+1) = cx_1(k) + (1-c)x_1(k-\tau(k)) \end{cases} \quad (78)$$

where $c \in [0, 1]$ is a constant, and $\tau(k)$ is a time-varying delay satisfying $d_1 \leq \tau(k) \leq d_2$.

Let $\theta(k) = cx_1(k) + (1-c)x_1(k-\tau(k)) \in [-m, m]$, where $m > 0$. Similar to [26], the T-S fuzzy model of (78) can be given as follows.

Plant Rule 1: IF $\theta(k)$ is $-m$, THEN

$$x(k+1) = A_1x(k) + A_{d1}x(k-\tau(k)) + B_1u^*(k).$$

Plant Rule 2: IF $\theta(k)$ is m , THEN

$$x(k+1) = A_2x(k) + A_{d2}x(k-\tau(k)) + B_2u^*(k)$$

TABLE I
COMPARISON OF THE MINIMAL H_∞ PERFORMANCE LEVEL γ

Methods	$d_2 = 14$	$d_2 = 15$	$d_2 = 20$	$d_2 = 25$
Theorem 3 [30]	0.8369	-	-	-
Theorem 2 [25]	0.1246	0.1246	0.1246	0.1247
Proposition 2	0.1047	0.1065	0.1078	0.1104

where $u^*(k) = 1.4 + u(k)$, and

$$A_1 = \begin{bmatrix} cm & 0.3 \\ c & 0 \end{bmatrix}, A_{d1} = \begin{bmatrix} (1-c)m & 0 \\ 1-c & 0 \end{bmatrix}, B_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} -cm & 0.3 \\ c & 0 \end{bmatrix}, A_{d2} = \begin{bmatrix} -(1-c)m & 0 \\ 1-c & 0 \end{bmatrix}, B_2 = B_1.$$

For this example, the stabilization problem is studied by using T-S fuzzy state feedback control in [26]. Set $d_1 = 1$, $c = 0.8$, and $m = 2$. By [26, Th. 2], the maximum upper bound $d_{\max} = 4$ of d_2 is obtained such that the system is stabilizable under a state feedback fuzzy controller. However, applying Proposition 2 in this paper yields a larger upper bound $d_{\max} = 6$, and the corresponding state feedback gains is calculated as

$$K_1 = [-2.4450 \quad -0.1964] \quad K_2 = [0.6626 \quad -0.1993].$$

Under the obtained controller, the initial state responses of the closed-loop system with the initial value $\phi(k) = \text{col}\{e^{k/6}, 0\}$ ($k = -6, -5, \dots, 0$) are depicted in Fig. 1, where the control input is imposed on the system starting from $k = 30$. From this figure, one can see that the closed-loop system is asymptotically stable for $1 \leq \tau(t) \leq 6$.

Example 2: Consider the following uncertain nonlinear time-delay system which is borrowed from [30] and also adopted in [25]:

$$\begin{cases} x_1(k+1) = -x_1^2(k) + 0.1x_1(k)x_1(k-\tau(k)) \\ \quad - 0.2x_1(k)x_2(k-\tau(k)) + x_1(k)w(k) \\ \quad + 0.02 \cos(k)x_1(k)x_2(k) + 0.3x_2(k) \\ \quad + 0.01 \sin(k)x_2(k-\tau(k)) + u_1(k) \\ x_2(k+1) = 0.1x_1(k) + x_2(k) - 0.1x_1(k)x_2(k-\tau(k)) \\ \quad + 0.02 \cos(k)x_2(k) + 0.5u_2(k) \\ z(k) = 0.1x_1^2(k) + 0.2x_2(k-\tau(k)) + 0.1w(k) \end{cases} \quad (79)$$

where $\tau(k)$ is a time-varying delay satisfying $d_1 \leq \tau(k) \leq d_2$, and $w(k)$ is the disturbance input given by $w(k) = 1/(16 + 15k)$. Suppose that $x_1(k) \in [-0.5, 0.5]$ and the membership functions are selected as

$$h_1(x_1(k)) = \frac{1}{2}(1-2x_1(k)), \quad h_2(x_1(k)) = \frac{1}{2}(1+2x_1(k)).$$

Denote

$$x(k) = \text{col}\{x_1(k), x_2(k)\}, \quad u(k) = \text{col}\{u_1(k), u_2(k)\}.$$

Then the nonlinear system in (79) can be represented by the following T-S fuzzy model with two rules.

Plant Rule i: IF $\theta(k) = x_1(k)$ is $h_i(x_1(k))$, THEN

$$\begin{cases} x(k+1) = \tilde{A}_ix(k) + \tilde{A}_{di}x(k-\tau(k)) \\ \quad + B_{1i}u(k) + B_{2i}w(k) \\ x(k) = \phi(k), \quad k \in \{-d_2, -d_2+1, \dots, 0\} \\ z(k) = L_ix(k) + L_{di}x(k-\tau(k)) + B_{3i}w(k) \end{cases} \quad (80)$$

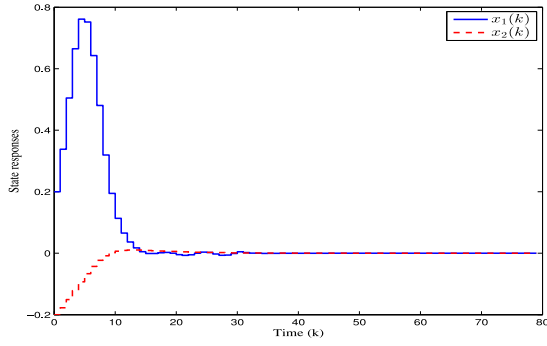
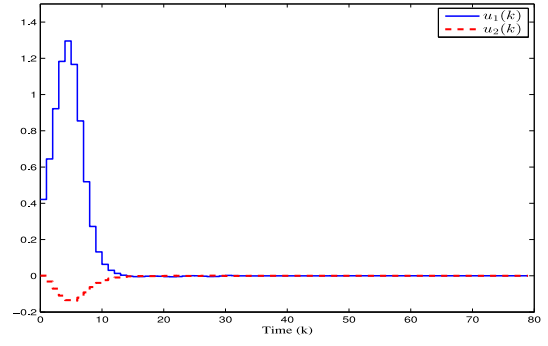
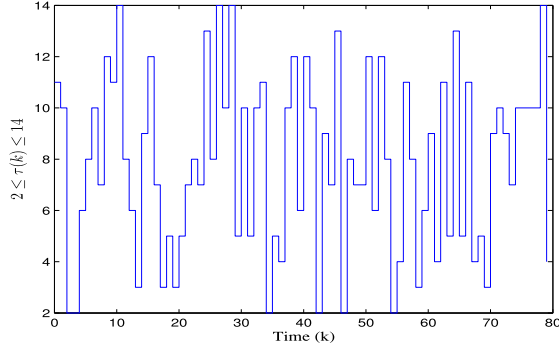


Fig. 2. State responses of the system (79) under the controller (81).

Fig. 4. Control input $u(k)$.Fig. 3. Time-varying delay $\tau(k)$ in Example 2.

where $i \in \mathcal{S} := \{1, 2\}$ and

$$\begin{aligned} \tilde{A}_i &= A_i + D_{1i}F_i(k)E_{1i}, \quad \tilde{A}_{di} = A_{di} + D_{1i}F_i(k)E_{2i} \\ A_1 &= \begin{bmatrix} 0.5 & 0.3 \\ 0.1 & 1.0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} -0.5 & 0.3 \\ 0.1 & 1.0 \end{bmatrix} \\ A_{d1} &= \begin{bmatrix} -0.05 & 0.1 \\ 0 & 0.05 \end{bmatrix}, \quad A_{d2} = \begin{bmatrix} 0.05 & -0.1 \\ -0 & -0.05 \end{bmatrix} \\ B_{11} &= B_{12} = \text{diag}\{1.0, 0.5\}, \quad B_{21} = \text{col}\{-0.5, 0\} \\ L_1 &= [-0.05 \ 0], \quad L_2 = -L_1, \quad B_{22} = -B_{21} \\ L_{d1} &= L_{d2} = [0 \ 0.2], \quad B_{31} = B_{32} = 0.1 \\ D_{11} &= \begin{bmatrix} -0.05 & 0.1 \\ 0.1 & 0 \end{bmatrix}, \quad D_{12} = \begin{bmatrix} 0.05 & 0.1 \\ 0.1 & 0 \end{bmatrix} \\ E_{11} &= E_{12} = \begin{bmatrix} 0 & 0.2 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = E_{22} = \text{diag}\{0, 0.1\} \\ F_1(k) &= \text{diag}\{\cos(k), \sin(k)\}, \quad F_2(k) = F_1(k). \end{aligned}$$

For the system (79) with an interval-like time-varying delay $2 \leq \tau(k) \leq d_2$, robust H_∞ control is investigated in [30] using a T-S fuzzy model approach. For $d_2 = 14$, the minimum H_∞ performance level $\gamma_{\min} = 0.8369$ is obtained under a certain H_∞ fuzzy controller. The same problem is also dealt with in [25], where a free-weighting matrix approach is proposed. For $d_2 = 14$, a smaller level $\gamma_{\min} = 0.1246$ is derived, which is only 14.89% of the one in [30]. It should be mentioned that, up to date, no any comparable results to $\gamma_{\min} = 0.1246$ are reported in the literature, which means that the result in [25] for this example is difficult to be improved. However, applying Proposition 2 to this example gives a smaller level $\gamma_{\min} = 0.1047$ for $d_2 = 14$, which is improved by 15.97%

if compared with [25]. Table I lists the achieved H_∞ performance levels $\gamma_{\min}$ for different values of d_2 using the methods in [25], [30], and this paper. From this table, one can see clearly that this paper presents some smaller H_∞ performance levels $\gamma_{\min}$ than [25] and [30], which is due to the fact that two new finite-sum inequalities and the matrix-based quadratic convex approach are introduced. Moreover, for $d_2 = 14$ and $\gamma = 0.12$, the H_∞ fuzzy controller can be calculated by

$$\begin{aligned} u(k) &= 0.5[(1-2x_1(k))K_1 + (1+2x_1(k))K_2]x(k) \\ K_1 &= \begin{bmatrix} 0.0585 & 1.9298 \\ -0.2000 & -0.2194 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0.9640 & -2.8486 \\ -0.2000 & -0.1989 \end{bmatrix}. \end{aligned} \quad (81)$$

With the controller (81), the state responses of the system (79) are plotted in Fig. 2, where the initial condition is given by $\phi(k) = \text{col}\{0.02, -0.02\}$ and the time-varying delay $\tau(k)$ is generated randomly between 2 and 14 as shown in Fig. 3. Fig. 4 depicts the corresponding control signal $u(k)$. From Figs. 2 and 4, one can see that within 10 s, the system state approaches its equilibrium point under the controller with relatively small control effort. After 10 s, the closed-loop system reaches its equilibrium state and the control effort goes down to zero. However, the methods proposed in [25] and [30] fail to draw any conclusion on the fuzzy H_∞ control problem of the system (79) with $d_2 = 14$ for $\gamma = 0.12$.

On the other hand, from Table I, one can see that for $\gamma = 0.1246$, using the approach in [25], the obtained maximum upper bound $d_{2\max}$ of d_2 is less than 25. Nevertheless, applying Proposition 2 yields $d_{2\max} = 29$, which is larger than the one in [25].

To further demonstrate the effectiveness of the proposed method, we consider the backing-up control of a computer-simulated truck-trailer, which is borrowed from [2] and [51].

Example 3: Consider the backing-up control of a computer-simulated truck-trailer shown in Fig. 5, where the time-delay model of the truck-trailer system with a disturbance input is given by (82) as shown at the bottom of the next page, where the model parameters are given as $v = -1.0$, $t = 2.0$, $\rho = 5.5$, and $\varpi = 2.8$. The time-varying delay $\tau(k)$ satisfies $d_1 \leq \tau(k) \leq d_2$. The constant κ is the retarded coefficient, which belongs to $[0, 1]$. In particular, for $\kappa = 0$, the truck-trailer system is related closely to the delay state $x_1(k - d(k))$ not to the instant state $x_1(k)$; while for $\kappa = 1$, it reduces to the system without time-delay, which is studied in [51]. In the sequel, we

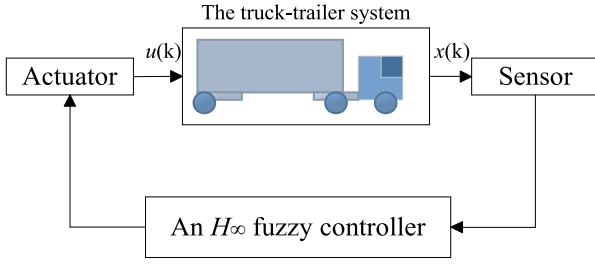


Fig. 5. Block diagram of H_∞ fuzzy control of a computer-simulated truck-trailer system.

set $\kappa = 0.65$. Then, both first and second equations in (82) are linear on the states $x_1(k)$ and $x_2(k)$, while the third is nonlinear on them. Therefore, the following fuzzy rules are employed.

Plant Rule 1: IF

$$\theta(k) = x_2(k) + \frac{\kappa vt}{2\rho} x_1(k) + \frac{(1-\kappa)vt}{2\rho} x_1(k-\tau(k)) \text{ is about } 0$$

THEN

$$x(k+1) = \tilde{A}_1 x(k) + \tilde{A}_{d1} x(k-\tau(k)) + B_{11} u(k) + B_{21} w(k).$$

Plant Rule 2: IF

$$\theta(k) = x_2(k) + \frac{\kappa vt}{2\rho} x_1(k) + \frac{(1-\kappa)vt}{2\rho} x_1(k-\tau(k)) \text{ is } \pm \pi$$

THEN

$$x(k+1) = \tilde{A}_2 x(k) + \tilde{A}_{d2} x(k-\tau(k)) + B_{12} u(k) + B_{22} w(k) - vt \zeta \pi \cdot \text{sign}(\theta(k))$$

where $\tilde{A}_i = A_i + \Delta A_i$ and $\tilde{A}_{di} = A_{di} + \Delta A_{di}$, $i = 1, 2$ with

$$A_1 = \begin{bmatrix} \kappa - \frac{\kappa vt}{\rho} & 0 & 0 \\ \frac{\kappa vt}{\rho} & 1 & 0 \\ \frac{\kappa v^2 t^2}{2\rho} & vt & 1 \end{bmatrix}, A_{d1} = \begin{bmatrix} 1 - \kappa - \frac{(1-\kappa)vt}{\rho} & 0 & 0 \\ \frac{(1-\kappa)vt}{\rho} & 0 & 0 \\ \frac{(1-\kappa)v^2 t^2}{2\rho} & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \kappa - \frac{\kappa vt}{\rho} & 0 & 0 \\ \frac{\kappa vt}{\rho} & 1 & 0 \\ \frac{\kappa \zeta v^2 t^2}{2\rho} & \zeta vt & 1 \end{bmatrix}, A_{d2} = \begin{bmatrix} 1 - \kappa - \frac{(1-\kappa)vt}{\rho} & 0 & 0 \\ \frac{(1-\kappa)vt}{\rho} & 0 & 0 \\ \frac{(1-\kappa)\zeta v^2 t^2}{2\rho} & 0 & 0 \end{bmatrix}$$

$$B_{11} = B_{12} = \text{col}\left\{\frac{vt}{\omega}, 0, 0\right\}, B_{21} = B_{22} = \text{col}\{1, 0, 0\}$$

and $\zeta \pi \cdot \text{sign}(\theta(k))$ denotes the offset in the second model. Similar to [2] and [51], we set $\zeta = 0.01/\pi$. Then the membership functions can be given by

$$h_1(\theta(k)) = \frac{\sin(\theta(k))}{\theta(k)}, \quad h_2(\theta(k)) = 1 - h_1(\theta(k)).$$

In the above uncertain fuzzy models, the modeling error uncertainty is assumed to be of form

$$\Delta A_i = D_{1i} F_i(k) E_{1i}, \quad \Delta A_{di} = D_{1i} F_i(k) E_{2i}$$

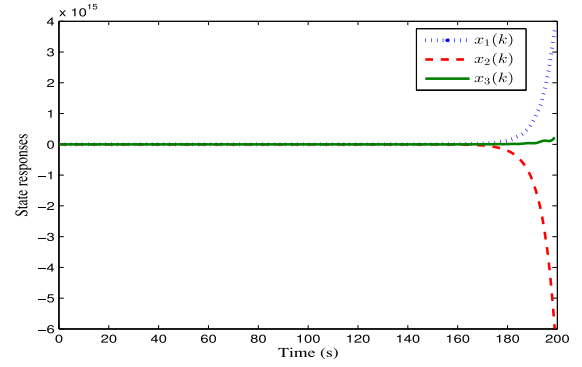


Fig. 6. State responses of the truck-trailer system (82) with $u(k) \equiv 0$.

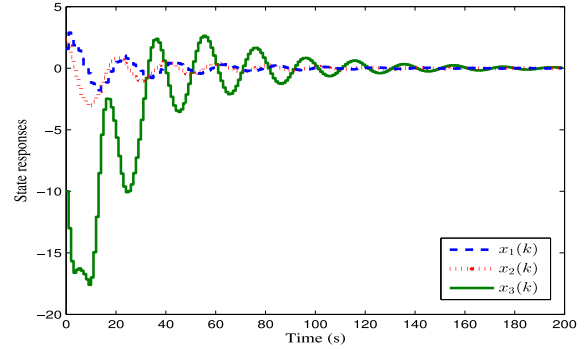


Fig. 7. State responses of the closed-loop truck-trailer system (82) under the controller (10) with (83).

where $F_i^T(k) F_i(k) \leq I$ ($i = 1, 2$), $D_{11} = D_{12} = \text{col}\{0, 0, 0.01\}$, $E_{11} = E_{12} = [(\kappa v^2 t^2 / 2\rho) \ 1 \ 0]$, and $E_{21} = E_{22} = [(((1-\kappa)\zeta v^2 t^2) / 2\rho) \ \zeta \ 0]$. Suppose that there is no uncertainty in the controlled output. Thus, $D_{21} = D_{22} = 0$, $L_1 = L_2 = [-0.1 \ 0.2 \ 0.1]$, $L_{d1} = L_{d2} = [0.5 \ -0.3 \ 0.2]$, and $B_{31} = B_{32} = 0$. The initial condition of the system is set to be $\phi(k) = \text{col}\{0.5\pi, 0.75\pi, -10\}$.

Let $d_1 = 2$ and $d_2 = 3$. Then, the initial condition responses of the truck-trailer system (82) with $u(k) \equiv 0$ are depicted in Fig. 6, from which, it is clear that the open loop of the truck-trailer system (82) is unstable. However, applying Proposition 2, it is found that the robust H_∞ control problem is feasible. In fact, if setting the tuning parameter $\varrho = 1.5$, the minimum H_∞ performance level $\gamma_{\min} = 0.4271$ is derived and the corresponding H_∞ fuzzy controller can be designed as (10) with

$$\begin{cases} K_1 = [1.4587 \ -1.217 \ 0.0610] \\ K_2 = [1.4107 \ -0.9393 \ 0.0607]. \end{cases} \quad (83)$$

Connecting this controller with the system (82), the state responses of the closed-loop truck-trailer system are illustrated

$$\begin{cases} x_1(k+1) = \left(\kappa - \frac{\kappa vt}{\rho}\right) x_1(k) + \left(1 - \kappa - \frac{(1-\kappa)vt}{\rho}\right) x_1(k-\tau(k)) + \frac{vt}{\omega} u(k) + w(k) \\ x_2(k+1) = x_2(k) + \frac{\kappa vt}{\rho} x_1(k) + \frac{(1-\kappa)vt}{\rho} x_1(k-\tau(k)) \\ x_3(k+1) = x_3(k) + vt \sin\left(x_2(k) + \frac{\kappa vt}{2\rho} x_1(k) + \frac{(1-\kappa)vt}{2\rho} x_1(k-\tau(k))\right) \end{cases} \quad (82)$$

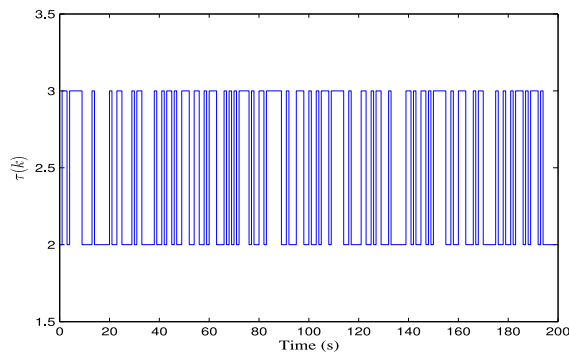


Fig. 8. Time-varying delay $\tau(k)$ in Example 3.

in Fig. 7. In the simulation, the time-varying delay $\tau(k)$ is generated randomly within its range, as is shown in Fig. 8. One can see clearly that from Fig. 7, the truck-trailer system (82) is stabilizable in the sense of H_∞ , which demonstrates the effectiveness of the proposed H_∞ fuzzy control method.

VII. CONCLUSION

The problem of robust H_∞ control for discrete-time T-S fuzzy systems with interval-like time-varying delays has been addressed. A novel finite-sum inequality-based method has been proposed to provide a tighter estimation on the forward difference of the FBD Lyapunov functional. Two novel inequalities have been established to provide a tighter bound for finite-sum terms appearing in the forward difference of the Lyapunov functional. The idea of a matrix-based quadratic convex approach has been applied to formulate some less conservative BRL for the resultant closed-loop T-S fuzzy system. Based on the BRL, a parameter-dependent condition has been presented to design suitable fuzzy H_∞ controllers, whose effectiveness has been demonstrated through two numerical examples and a truck-trailer system. The proposed method can be employed to deal with fuzzy control for networked systems with or without event-triggered schemes, which motivates our future research.

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Qing-Long Han (M'09–SM'13) received the B.Sc. degree in mathematics from Shandong Normal University, Jinan, China, in 1983, and the M.Sc. and Ph.D. degrees in control engineering and electrical engineering from the East China University of Science and Technology, Shanghai, China, in 1992 and 1997, respectively.

From 1997 to 1998, he was a Post-Doctoral Researcher Fellow with the Laboratoire d'Automatique et d'Informatique Industrielle (currently, Laboratoire d'Informatique et d'Automatique pour les Systèmes), École Supérieure d'Ingénieurs de Poitiers (currently, École Nationale Supérieure d'Ingénieurs de Poitiers), Université de Poitiers, Poitiers, France. From 1999 to 2001, he was a Research Assistant Professor with the Department of Mechanical and Industrial Engineering, Southern Illinois University at Edwardsville, Edwardsville, IL, USA. From 2001 to 2014, he was a Laureate Professor, the Associate Dean (Research and Innovation) with the Higher Education Division, and the Founding Director of the Centre for Intelligent and Networked Systems at Central Queensland University, Rockhampton, QLD, Australia. From 2014 to 2016, he was the Deputy Dean (Research) with the Griffith Sciences, and a Professor with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. In 2016, he joined the Swinburne University of Technology, Melbourne, VIC, Australia, where he is currently the Pro Vice-Chancellor (Research Quality) and a Distinguished Professor. In 2010, he was appointed as a Chang Jiang (Yangtze River) Scholar Chair Professor by the Ministry of Education of China. His current research interests include networked control systems, neural networks, time-delay systems, multiagent systems, and complex dynamical systems.

Prof. Han was a recipient of one of the World's Most Influential Scientific Minds from 2014–2016 and the Highly Cited Researcher Award in Engineering by Thomson Reuters. He is an Associate Editor of a number of international journals, including the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, the IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, the IEEE TRANSACTIONS ON CYBERNETICS, and *Information Sciences*.



Xian-Ming Zhang (M'16) received the M.Sc. degree in applied mathematics and the Ph.D. degree in control theory and engineering from Central South University, Changsha, China, in 1992 and 2006, respectively.

In 1992, he joined Central South University, where he was an Associate Professor with the School of Mathematics and Statistics. From 2007 to 2014, he was a Post-Doctoral Research Fellow and a Lecturer with the School of Engineering and Technology, Central Queensland University,

Rockhampton, QLD, Australia. From 2014 to 2016, he was a Lecturer with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. In 2016, he joined the Swinburne University of Technology, Melbourne, VIC, Australia, where he is currently a Senior Lecturer with the School of Software and Electrical Engineering. His current research interests include H_∞ filtering, event-triggered control systems, networked control systems, neural networks, distributed systems, and time-delay systems.

Dr. Zhang was a recipient of the National Natural Science Award (Level 2) in China in 2013, the Hunan Provincial Natural Science Award (Level 1) in Hunan Province, China, in 2011, and the IET Premium Award in 2016. He serves as an Associate Editor for the *Journal of the Franklin Institute* and an Editorial Board Member of *Neural Computing and Applications*.



Xiaohua Ge received the B.Eng. degree in electronic and information engineering from Nanchang Hangkong University, Nanchang, China, in 2008, the M.Eng. degree in control theory and control engineering from Hangzhou Dianzi University, Hangzhou, China, in 2011, and the Ph.D. degree in computer engineering from Central Queensland University, Rockhampton, QLD, Australia, in 2014.

He was a Research Assistant with the Centre for Intelligent and Networked Systems, Central Queensland University, from 2011 to 2013, where

he was also a Research Fellow with the Centre for Intelligent and Networked Systems, in 2014. From 2015 to 2016, he was a Research Fellow with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. He is currently a Lecturer with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC, Australia. His current research interests include networked control and filtering, distributed networked control systems, multiagent systems, and sensor networks.